Supersymmetry and Supergravity

沈黙の魔女の魔法講座

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Preface

Supersymmetry (SUSY for short) is a symmetry that relates particles of different spin: it turns bosons into fermions, and fermions into bosons. That sounds simple enough, but to really understand it you first have to answer two questions that look almost childish: what is "a kind of particle"? And what is a particle's "spin"? In nonrelativistic quantum mechanics spin is put in by hand, as an "intrinsic angular momentum". In relativity it has a deeper and much cleaner origin: the symmetry of spacetime itself. The first seven chapters of this book are devoted to making that statement precise, and to assembling every tool we will need later to build supersymmetric theories.

I assume you have had a semester of quantum mechanics, know the basics of special relativity, and are comfortable multiplying matrices, taking partial derivatives, and handling complex numbers. If words like group theory, Lie algebra, or spinor mean nothing to you, don't worry; each one is explained the first time it is used. I have tried not to skip steps: wherever I say "it is easy to verify", it really is easy, and I hope you will actually verify it.

The plan of the book is simple. Chapter 1 studies the symmetries of spacetime, namely the Lorentz and Poincaré groups. Chapter 2 asks how these groups can act on various objects, and this is where spinors appear, quite naturally. Chapter 3 uses those results to classify particles; you will see mass and spin (or, for massless particles, helicity) emerge as "invariant labels". Chapters 4 through 7 turn these abstract representations into concrete fields, namely scalar, spinor and vector fields, and along the way introduce the Grassmann numbers that fermion fields require, as well as Majorana spinors. By the end you should be holding a "ledger of degrees of freedom", and supersymmetry demands precisely that the bosonic and fermionic entries in that ledger balance.

Worked examples are scattered through the text; they show you how to put the formulas to work. Most sections end with problems. I suggest you do every problem marked with a *; those marked ** are harder, for readers who want to nail down the details. Solutions to all the problems are collected in Appendix B, but please try them yourself first. Reading someone else's solution and producing your own are two entirely different experiences.

A final word about notation. What gives beginners the most grief in this subject is usually not the physics but the signs: is the metric $(+,-,-,-)$ or $(-,+,+,+)$? Do the generators carry $+\mathrm{i}$ or $-\mathrm{i}$? Is ε^{12} equal to $+1$ or -1? Every book makes its own choices, and each choice ripples through everything else. This book uses one set of conventions from beginning to end, collected in Appendix A. When you compare a formula here with one from another book, check the conventions first and compare the results second. The best advice I can give you is this: keep a sheet of paper at hand, and work out every sign for yourself.

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Chapter 1

Spacetime Symmetry

The laws of physics should not depend on where you stand, which way you face, or how fast you are moving (uniformly). That is an old saying, but it packs far more punch than it seems to. In this chapter we translate "change the observer, and the physics stays the same" into precise mathematics: first we write down every allowed coordinate transformation, then we find their "generators", and finally we compute the commutation relations among those generators. That set of commutation relations is called the Poincaré algebra, and everything else in this book (the classification of particles, spinors, the various fields) is built on top of it.

1.1 Minkowski Spacetime

To pin down an event you need one time and three space coordinates. We will use natural units, $c = \hbar = 1$, throughout, and lump them together as

$$x^\mu = (x^0, x^1, x^2, x^3) = (t, x, y, z), \qquad \mu = 0, 1, 2, 3. \tag{1.1}$$

The superscript μ labels the component; it is not a power. This is easy to confuse at first: x^2 is the second spatial component y, whereas $(x^2)^2$ is y^2. By convention, Greek letters $\mu, \nu, \rho, \ldots$ run over $0, 1, 2, 3$, while Latin letters i, j, k run only over the spatial components $1, 2, 3$.

This four-dimensional space differs from ordinary Euclidean space in one essential respect: the "inner product" of two vectors is not obtained by multiplying matching components and adding them all up. Instead,

$$x \cdot y = x^0 y^0 - x^1 y^1 - x^2 y^2 - x^3 y^3. \tag{1.2}$$

The time and space parts enter with opposite signs. To keep track of that minus sign compactly, we introduce the **metric**

$$\eta_{\mu\nu} = \mathrm{diag}(1, -1, -1, -1), \tag{1.3}$$

so that $x \cdot y = \eta_{\mu\nu} x^\mu y^\nu$. Here I am using the Einstein summation convention: an index that appears twice in the same term, once up and once down, is automatically summed over. So $\eta_{\mu\nu} x^\mu y^\nu$ is really a sum of sixteen terms (only four of which are nonzero). A four-dimensional space equipped with this inner product is called **Minkowski spacetime**, written $\mathbb{R}^{1,3}$: one time direction, three space directions.

1.1.1 Upper and Lower Indices

With a metric in hand, we can "lower" an index:

$$x_\mu \equiv \eta_{\mu\nu} x^\nu, \qquad \text{that is,} \qquad x_\mu = (t, -x, -y, -z). \tag{1.4}$$

Conversely, $\eta^{\mu\nu}$ (numerically the same as $\eta_{\mu\nu}$) raises it again: $x^\mu = \eta^{\mu\nu} x_\nu$. Lowering an index changes nothing physical; it is just a different way of keeping the books. The payoff is that inner products can be written very cleanly:

$$x \cdot y = x^\mu y_\mu = x_\mu y^\mu, \qquad x^2 \equiv x^\mu x_\mu = t^2 - x^2 - y^2 - z^2. \tag{1.5}$$

(In that last expression x^2, y^2 are powers. Please forgive this unfortunate notation; luckily, the context always makes it clear.)

Partial derivatives come with a lower index built in:

$$\partial_\mu \equiv \frac{\partial}{\partial x^\mu} = (\partial_t, \partial_x, \partial_y, \partial_z), \qquad \partial^\mu = \eta^{\mu\nu}\partial_\nu = (\partial_t, -\partial_x, -\partial_y, -\partial_z). \tag{1.6}$$

Two little formulas follow immediately, and we will use them over and over:

$$\partial_\mu x^\nu = \delta_\mu{}^\nu, \qquad \partial_\mu x_\nu = \eta_{\mu\nu}. \tag{1.7}$$

The first says "the derivative of a coordinate with respect to itself is 1, with respect to any other coordinate it is 0"; the second just lowers one more index.

Example 1.1 The inner product is not Euclidean length

Let $x^\mu = (5, 3, 0, 0)$. Find x^2.

Solution Lowering the index gives $x_\mu = (5, -3, 0, 0)$, so $x^2 = x^\mu x_\mu = 25 - 9 = 16$. If you simply add up the squares of the four components you get 34. That is the square of the Euclidean length, and in relativity it is meaningless: switch to another reference frame and it changes. □

1.1.2 Intervals and Mass

The "length squared" $\Delta s^2 = \eta_{\mu\nu}\Delta x^\mu \Delta x^\nu$ of the separation Δx^μ between two events is called the interval. If $\Delta s^2 > 0$ the interval is timelike, and a massive object can pass through both events in turn; if $\Delta s^2 = 0$ it is lightlike, and light travels along such paths; if $\Delta s^2 < 0$ it is spacelike, and no signal of any kind can pass between the two events.

The four-momentum $p^\mu = (E, \boldsymbol{p})$ is also a four-vector, and its "length squared" is

$$p^2 = E^2 - \boldsymbol{p}^2 = m^2. \tag{1.8}$$

Different observers measure different E and $\boldsymbol{p}$, but they all get the same m^2. Hang on to this fact: it hints that mass is a quantity we can use to "label" particles. Chapter 3 will develop this idea into a complete classification.

Problem 1.1 Given $x^\mu = (2, 1, 0, 0)$, write down x_μ and x^2. Then compute $\partial_1 x_1$, and explain why the answer is not $+1$.

1.2 Lorentz Transformations

Now we ask: which linear transformations

$$x'^{\mu} = \Lambda^{\mu}{}_{\nu} x^{\nu} \tag{1.9}$$

leave the inner product unchanged? Plug in and demand that $x'^2 = x^2$ for every x:

$$\eta_{\mu\nu} x'^{\mu} x'^{\nu} = \eta_{\mu\nu} \Lambda^{\mu}{}_{\rho} \Lambda^{\nu}{}_{\sigma}\, x^{\rho} x^{\sigma} \overset{!}{=} \eta_{\rho\sigma} x^{\rho} x^{\sigma}. \tag{1.10}$$

Both sides are quadratic forms in x, and the coefficients are symmetric in ρ, σ, so the coefficients must match term by term:

$$\boxed{\eta_{\mu\nu} \Lambda^{\mu}{}_{\rho} \Lambda^{\nu}{}_{\sigma} = \eta_{\rho\sigma}} \qquad \text{or, as matrices,} \qquad \boxed{\Lambda^{\mathsf{T}} \eta \Lambda = \eta.} \tag{1.11}$$

This is the definition of a **Lorentz transformation**. The set of all real matrices satisfying it forms a group, called $O(1,3)$. To say it is a "group" just means this: two Lorentz transformations in succession give another Lorentz transformation; doing nothing (the identity) counts as one; and every transformation can be undone.

Taking the determinant of Eq. (1.11) gives $(\det \Lambda)^2 = 1$; looking at its 00 component gives $(\Lambda^0{}_0)^2 = 1 + \sum_i (\Lambda^i{}_0)^2 \geq 1$. So $O(1,3)$ falls into four pieces, according to whether $\det \Lambda = \pm 1$ and whether $\Lambda^0{}_0 \geq 1$ or ≤ -1. In this book we care almost exclusively about the piece that contains the identity,

$$\mathrm{SO}^+(1,3) = \{\Lambda \mid \det \Lambda = 1,\ \Lambda^0{}_0 \geq 1\}, \tag{1.12}$$

which consists of all rotations, all boosts (switching to a uniformly moving frame), and their combinations.[1]

1.2.1 Two Examples

A rotation by angle θ about the z axis:

$$\Lambda_R(\theta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{1.13}$$

It leaves t and z alone and preserves $x^2 + y^2$, so of course it preserves $t^2 - x^2 - y^2 - z^2$.

A boost along z is more interesting, because it mixes time and space:

$$\Lambda_B(\chi) = \begin{pmatrix} \cosh\chi & 0 & 0 & \sinh\chi \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \sinh\chi & 0 & 0 & \cosh\chi \end{pmatrix}. \tag{1.14}$$

The parameter χ is called the **rapidity**; it is related to the velocity by $\tanh\chi = v$, so that $\cosh\chi = \gamma = 1/\sqrt{1-v^2}$. Let's check that it really does preserve the interval:

$$\begin{aligned} t'^2 - z'^2 &= (t\cosh\chi + z\sinh\chi)^2 - (t\sinh\chi + z\cosh\chi)^2 \\ &= (\cosh^2\chi - \sinh^2\chi)(t^2 - z^2) = t^2 - z^2. \end{aligned} \tag{1.15}$$

[1] The other three pieces contain the spatial reflection $\boldsymbol{x} \to -\boldsymbol{x}$, time reversal $t \to -t$, and the combination of the two. None of them can be reached continuously from the identity, so they never show up among the generators we build below from "infinitesimal transformations".

The cross terms $2tz\cosh\chi\sinh\chi$ cancel exactly. Hyperbolic functions play the same role here that trigonometric functions play for rotations; the minus sign in the metric simply turns $\cos^2+\sin^2=1$ into $\cosh^2-\sinh^2=1$.

Rapidity is far more convenient than velocity: for two boosts in the same direction the rapidities simply add (Problem 1.3), whereas velocities need that famous relativistic addition formula. But there is an important difference between the two kinds of parameter worth noting right now: an angle comes back to where it started after one full turn, while rapidity can grow without bound. This "no end in sight" property is called **noncompactness**, and in Chapter 3 it will have real consequences.

1.2.2 Six Parameters

A real 4×4 matrix has 16 entries. Both sides of condition (1.11) are symmetric matrices, so it imposes $4\times5/2=10$ independent equations, leaving $16-10=6$ free parameters. Three rotations and three boosts: exactly six.

A more illuminating way to count is to look at a transformation very close to the identity,

$$\Lambda^\mu{}_\nu=\delta^\mu{}_\nu+\omega^\mu{}_\nu+O(\omega^2), \tag{1.16}$$

where ω is small. Substitute into Eq. (1.11) and keep only first-order terms:

$$\eta_{\rho\sigma}=\eta_{\mu\nu}(\delta^\mu{}_\rho+\omega^\mu{}_\rho)(\delta^\nu{}_\sigma+\omega^\nu{}_\sigma)=\eta_{\rho\sigma}+\omega_{\sigma\rho}+\omega_{\rho\sigma}+O(\omega^2), \tag{1.17}$$

where $\omega_{\rho\sigma}\equiv\eta_{\rho\mu}\omega^\mu{}_\sigma$. Therefore

$$\boxed{\omega_{\mu\nu}=-\omega_{\nu\mu}.} \tag{1.18}$$

An infinitesimal Lorentz transformation is described by a 4×4 antisymmetric matrix. The diagonal elements of an antisymmetric matrix vanish, and the independent off-diagonal elements are the six $(01),(02),(03),(12),(23),(31)$: the first three are boosts, the last three rotations. In d-dimensional spacetime the same argument gives $d(d-1)/2$ parameters.

***Problem 1.2** In the t,z plane take $t=2$, $z=1$. Using Eq. (1.14) with $v=\tanh\chi=3/5$, find t',z', and check that the interval is unchanged.

Problem 1.3 Show that $\Lambda_B(\chi_1)\Lambda_B(\chi_2)=\Lambda_B(\chi_1+\chi_2)$. Use this to derive the relativistic velocity addition formula $v=(v_1+v_2)/(1+v_1v_2)$.

1.3 Translations and the Poincaré Group

Besides rotations and boosts, shifting the origin of coordinates obviously doesn't change the physics either. The most general interval-preserving transformation is

$$x'^\mu=\Lambda^\mu{}_\nu x^\nu+a^\mu, \tag{1.19}$$

where a^μ is a constant four-vector: a^0 shifts the origin of time, a^i the origin of space. A translation doesn't change the difference between two events, so it doesn't change the interval

either. Together these transformations form the **Poincaré group**, with $6 + 4 = 10$ parameters in all.

The multiplication law of the Poincaré group has a small twist. Do (Λ_2, a_2) first, then (Λ_1, a_1):

$$x'' = \Lambda_1(\Lambda_2 x + a_2) + a_1 = (\Lambda_1\Lambda_2)\, x + (a_1 + \Lambda_1 a_2), \tag{1.20}$$

so

$$(\Lambda_1, a_1)(\Lambda_2, a_2) = (\Lambda_1\Lambda_2,\ a_1 + \Lambda_1 a_2). \tag{1.21}$$

If the Lorentz part and the translation part had nothing to do with each other, the second entry would be $a_1 + a_2$. Instead it is $a_1 + \Lambda_1 a_2$, because the Lorentz transformation done second also rotates the direction of the translation done first. Imagine walking ten steps east, after which the whole coordinate system turns ninety degrees to the left: in the new coordinates, those ten steps point "north". Mathematicians call this kind of structure a **semidirect product**, and write

$$\mathbb{R}^{1,3} \rtimes \mathrm{SO}^+(1,3). \tag{1.22}$$

Problem 1.4 Use Eq. (1.21) to find the inverse of (Λ, a).

1.4 Generators

Continuous groups have one enormous advantage: every finite transformation can be built up from many infinitesimal ones, so it is enough to understand what happens when you "move just a tiny bit". The thing that describes that tiny bit is called a **generator**.

The simplest example is translation. Taylor-expand any smooth function:

$$f(x + a) = f(x) + a^\mu \partial_\mu f(x) + \frac{1}{2} a^\mu a^\nu \partial_\mu \partial_\nu f(x) + \cdots = \exp(a^\mu \partial_\mu)\, f(x). \tag{1.23}$$

The partial derivative "generates" translations: compose the infinitesimal translation $1 + a \cdot \partial$ infinitely many times and you get the exponential. Physicists like to put an i into the definition of generators, so that in quantum mechanics they are Hermitian operators, corresponding to observables. In this book we take

$$\boxed{P_\mu = \mathrm{i}\partial_\mu, \qquad L_{\mu\nu} = \mathrm{i}(x_\mu \partial_\nu - x_\nu \partial_\mu) = -L_{\nu\mu}.} \tag{1.24}$$

Then $e^{-\mathrm{i}a\cdot P} f(x) = f(x + a)$. P_μ is just the four-momentum operator you know from quantum mechanics: a plane wave $e^{-\mathrm{i}p\cdot x}$ satisfies $P_\mu e^{-\mathrm{i}p\cdot x} = p_\mu e^{-\mathrm{i}p\cdot x}$.[2]

What is $L_{\mu\nu}$? One component tells the story. Since $x_1 = -x$ and $x_2 = -y$,

$$L_{12} = \mathrm{i}(x_1\partial_2 - x_2\partial_1) = -\mathrm{i}(x\partial_y - y\partial_x). \tag{1.25}$$

[2] Many books take $P_\mu = -\mathrm{i}\partial_\mu$ and $L_{\mu\nu} = -\mathrm{i}(x_\mu\partial_\nu - x_\nu\partial_\mu)$. That is neither right nor wrong in itself, but then the commutation relations of the next section, the signs in the exponents of finite transformations, and the names "left-handed" and "right-handed" for spinors all have to change along with it. Never copy one formula from one book and another formula from another and stitch them together.

That is precisely the orbital angular momentum L_z of quantum mechanics. The L_{ij} generate spatial rotations, the L_{0i} generate boosts. A general Lorentz transformation (connected to the identity) can be written

$$D(\omega) = \exp\Big(-\frac{\mathrm{i}}{2}\omega^{\mu\nu}L_{\mu\nu}\Big) = \mathbf{1} - \frac{\mathrm{i}}{2}\omega^{\mu\nu}L_{\mu\nu} + O(\omega^2). \tag{1.26}$$

The $\frac{1}{2}$ is there because $\omega^{\mu\nu}$ and $L_{\mu\nu}$ are both antisymmetric, so when we sum over all μ, ν each pair of indices gets counted twice ($\omega^{01}L_{01}$ and $\omega^{10}L_{10}$ are equal).

1.5 The Poincaré Algebra

Does the order of two transformations matter? For translations, no: going east then north lands you at the same place as going north then east. For rotations it does: take a book, rotate it ninety degrees about a vertical axis and then ninety degrees about a horizontal axis, and compare with doing it the other way around. The results are not the same. At the infinitesimal level, this order dependence is captured exactly by the **commutator** of the generators, $[A, B] = AB - BA$.

When computing commutators of differential operators there is one trick, and it is also the step beginners most often miss: let the operators act on an arbitrary "test function" $f(x)$. For example,

$$[\partial_\mu, x_\nu]f = \partial_\mu(x_\nu f) - x_\nu\partial_\mu f = (\partial_\mu x_\nu)f = \eta_{\mu\nu}f, \tag{1.27}$$

so $[\partial_\mu, x_\nu] = \eta_{\mu\nu}$. All we used was the product rule and Eq. (1.7). Every calculation below is a repetition of this one move.

1.5.1 Translation with Translation, Lorentz with Translation

Partial derivatives commute with one another, so

$$\boxed{[P_\mu, P_\nu] = 0.} \tag{1.28}$$

Next, L with P. Using a test function we first find $[x_\mu\partial_\nu, \partial_\rho] = -\eta_{\mu\rho}\partial_\nu$, and then

$$\begin{aligned}[L_{\mu\nu}, P_\rho] &= \mathrm{i}\cdot\mathrm{i}\,[x_\mu\partial_\nu - x_\nu\partial_\mu, \partial_\rho] = \eta_{\mu\rho}\partial_\nu - \eta_{\nu\rho}\partial_\mu \\ &= \mathrm{i}\big(\eta_{\nu\rho}P_\mu - \eta_{\mu\rho}P_\nu\big),\end{aligned} \tag{1.29}$$

where in the last step we substituted back $\partial_\mu = -\mathrm{i}P_\mu$. What this result says is that a Lorentz transformation mixes the four components of P_μ, in exactly the way it mixes the components of any four-vector. In other words, four-momentum is a four-vector. No surprise there, but now we have actually computed it.

1.5.2 Lorentz with Lorentz

This one is a bit longer, but there is nothing new in it. Write $A_{\mu\nu} = x_\mu\partial_\nu - x_\nu\partial_\mu$, so that $L = \mathrm{i}A$. First compute a building block:

$$[x_\mu\partial_\nu, x_\rho\partial_\sigma]f = x_\mu\partial_\nu(x_\rho\partial_\sigma f) - x_\rho\partial_\sigma(x_\mu\partial_\nu f) = \eta_{\nu\rho}\,x_\mu\partial_\sigma f - \eta_{\sigma\mu}\,x_\rho\partial_\nu f. \tag{1.30}$$

The second-derivative term $x_\mu x_\rho\partial_\nu\partial_\sigma f$ is the same in both orders and drops out in the difference; what survives are the two terms where the product rule hits a coordinate. Now break $[A_{\mu\nu}, A_{\rho\sigma}]$

into four building blocks:

$$\begin{aligned}[A_{\mu\nu}, A_{\rho\sigma}] &= [x_\mu\partial_\nu, x_\rho\partial_\sigma] - [x_\mu\partial_\nu, x_\sigma\partial_\rho] - [x_\nu\partial_\mu, x_\rho\partial_\sigma] + [x_\nu\partial_\mu, x_\sigma\partial_\rho] \\ &= (\eta_{\nu\rho}x_\mu\partial_\sigma - \eta_{\sigma\mu}x_\rho\partial_\nu) - (\eta_{\nu\sigma}x_\mu\partial_\rho - \eta_{\rho\mu}x_\sigma\partial_\nu) \\ &\quad - (\eta_{\mu\rho}x_\nu\partial_\sigma - \eta_{\sigma\nu}x_\rho\partial_\mu) + (\eta_{\mu\sigma}x_\nu\partial_\rho - \eta_{\rho\nu}x_\sigma\partial_\mu). \end{aligned} \tag{1.31}$$

Group the eight terms in pairs according to their metric factor, and each pair assembles itself into an A. For example, the two terms carrying $\eta_{\nu\rho}$ are $\eta_{\nu\rho}(x_\mu\partial_\sigma - x_\sigma\partial_\mu) = \eta_{\nu\rho}A_{\mu\sigma}$. Putting the four pairs together:

$$[A_{\mu\nu}, A_{\rho\sigma}] = \eta_{\nu\rho}A_{\mu\sigma} - \eta_{\mu\rho}A_{\nu\sigma} - \eta_{\nu\sigma}A_{\mu\rho} + \eta_{\mu\sigma}A_{\nu\rho}. \tag{1.32}$$

Finally switch back to L: the left side picks up $\mathrm{i}^2 = -1$, and substituting $A = -\mathrm{i}L$ on the right brings in a $-\mathrm{i}$; together they make a single i:

$$\boxed{[L_{\mu\nu}, L_{\rho\sigma}] = \mathrm{i}\big(\eta_{\nu\rho}L_{\mu\sigma} - \eta_{\mu\rho}L_{\nu\sigma} - \eta_{\nu\sigma}L_{\mu\rho} + \eta_{\mu\sigma}L_{\nu\rho}\big).} \tag{1.33}$$

$[P, P] = 0$, Eq. (1.29), and Eq. (1.33) together constitute the **Poincaré algebra**. It looks like just three lines of formulas, yet it records, without omitting a single word, which continuous symmetry operations spacetime has and how they are entangled with one another.

1.5.3 One Algebra, Many Realizations

We derived the Poincaré algebra using the differential operators (1.24), acting on functions. But the algebra itself doesn't care what the generators look like in detail: any set of matrices or operators obeying the same commutation relations can be used to implement the same symmetry operations. Such a realization is called a **representation** of the algebra (or of the group).

Equation (1.24) describes only the part of the transformation that "moves a scalar function around as a whole"; this is called the orbital part. If a field has several components (the electric field has three, for instance), a rotation also mixes those components among themselves. The complete generator is then

$$L^{\text{total}}_{\mu\nu} = L^{\text{orbital}}_{\mu\nu} + \Sigma_{\mu\nu}, \tag{1.34}$$

where $\Sigma_{\mu\nu}$ is a constant matrix that acts only on the components. It commutes with L^{orbital}, and it satisfies Eq. (1.33) on its own. For a scalar field $\Sigma = 0$; for spinor and vector fields Σ is nonzero, and this is precisely the mathematical identity of "spin". The task of the next chapter is to find all possible Σ's.

***Problem 1.5** Using $L_{12} = -\mathrm{i}(x\partial_y - y\partial_x)$ and $P^1 = -\mathrm{i}\partial_x$ directly, act on a test function to compute $[L_{12}, P^1]f$, and compare with Eq. (1.29).

Chapter 2

Representations of the Lorentz Group and Spinors

At the end of the last chapter we saw that, for a field with components, a Lorentz transformation does more than "move the field around as a whole": the components themselves get mixed, and the way they mix is fixed by a set of constant matrices $\Sigma_{\mu\nu}$ obeying the Lorentz algebra. So what matrices of that kind are there? In mathematical language: what are the finite-dimensional representations of the Lorentz algebra?

At first hearing this sounds like a hard problem, but a beautiful trick turns it into one you already know well: the Lorentz algebra can be viewed as *two copies* of the angular momentum algebra. And you found every representation of the angular momentum algebra back in quantum mechanics. This chapter follows that road, and at the end of it an object will appear that you may never have met before: the spinor.

2.1 Rotations and Boosts

Of the six generators $L_{\mu\nu}$, three take care of rotations and three of boosts. Let us write them separately:

$$\boxed{R_i = \frac{1}{2}\varepsilon_{ijk}L_{jk}, \qquad K_i = L^{0i}, \qquad \varepsilon_{123} = 1.} \tag{2.1}$$

Here j, k are simply summed over $1, 2, 3$ in the ordinary way. Explicitly, $R_1 = L_{23}$, $R_2 = L_{31}$, $R_3 = L_{12}$; for example, $R_3 = \frac{1}{2}(\varepsilon_{312}L_{12} + \varepsilon_{321}L_{21}) = \frac{1}{2}(L_{12} - L_{21}) = L_{12}$. Since lowering two spatial indices brings in two minus signs, $L_{jk} = L^{jk}$; whereas K_i involves only one spatial index, so $L^{0i} = -L_{0i}$.[1]

R_3 is just the L_z we met in the last chapter, and it generates rotations about the z axis. K_3 generates boosts along z: using the vector representation of Chapter 1 you can check that $e^{-\mathrm{i}\theta R_3}$ is exactly the rotation matrix $\Lambda_R(\theta)$, and $e^{-\mathrm{i}\chi K_3}$ is exactly the $\Lambda_B(\chi)$ of Eq. (1.14).

Breaking Eq. (1.33) into components, we get

$$\boxed{[R_i, R_j] = \mathrm{i}\varepsilon_{ijk}R_k, \qquad [R_i, K_j] = \mathrm{i}\varepsilon_{ijk}K_k, \qquad [K_i, K_j] = -\mathrm{i}\varepsilon_{ijk}R_k.} \tag{2.2}$$

You can pick a component and check it yourself. For instance,

$$[K_1, K_2] = [L^{01}, L^{02}] = [L_{01}, L_{02}] = -\mathrm{i}L_{12} = -\mathrm{i}R_3,$$

where in the second equality the two minus signs multiply to a plus, and the third equality is read straight off Eq. (1.33).

[1] Some books take $K_i = L_{0i}$, which differs by an overall sign. That doesn't affect any of the commutation relations below, but it swaps the names of J^L and J^R later on, and with them the names "left-handed spinor" and "right-handed spinor". We choose ours so that a boost takes the form $e^{-\mathrm{i}\chi K_3}$, exactly parallel to a rotation.

Each of these three relations has a meaning. The first is the angular momentum algebra, no surprise there. The second says that $\boldsymbol{K}$ behaves under rotations like an ordinary three-vector: the direction of a boost turns along with the coordinate axes. The third is the most interesting: do two boosts in different directions one after the other, and the results for the two orders differ by a *rotation*. This is the origin of the famous Thomas precession of relativity. Pay special attention to that minus sign. If it were a plus, R and K together would form the rotation algebra of four-dimensional Euclidean space; it is precisely the minus sign in the metric that makes spacetime different from ordinary four-dimensional space.

2.2 Two Copies of Angular Momentum

In Eq. (2.2), R and K are tangled together, which makes them awkward to study directly. Try the following combinations:

$$J_i^L = \frac{1}{2}(R_i + \mathrm{i}K_i), \qquad J_i^R = \frac{1}{2}(R_i - \mathrm{i}K_i). \tag{2.3}$$

Conversely, $R_i = J_i^L + J_i^R$ and $K_i = -\mathrm{i}(J_i^L - J_i^R)$. Let's compute the commutator of J^L with itself:

$$\begin{aligned}[J_i^L, J_j^L] &= \frac{1}{4}\Big([R_i, R_j] + \mathrm{i}[R_i, K_j] + \mathrm{i}[K_i, R_j] - [K_i, K_j]\Big) \\ &= \frac{1}{4}\Big(\mathrm{i}\varepsilon_{ijk}R_k - \varepsilon_{ijk}K_k - \varepsilon_{ijk}K_k + \mathrm{i}\varepsilon_{ijk}R_k\Big) = \mathrm{i}\varepsilon_{ijk}\,\frac{1}{2}\big(R_k + \mathrm{i}K_k\big) = \mathrm{i}\varepsilon_{ijk}J_k^L.\end{aligned} \tag{2.4}$$

(For the third term we used $[K_i, R_j] = -[R_j, K_i] = -\mathrm{i}\varepsilon_{jik}K_k = \mathrm{i}\varepsilon_{ijk}K_k$.) The calculation for J^R is exactly the same. What about the mixed one?

$$[J_i^L, J_j^R] = \frac{1}{4}\Big([R_i, R_j] - \mathrm{i}[R_i, K_j] + \mathrm{i}[K_i, R_j] + [K_i, K_j]\Big) = 0, \tag{2.5}$$

since the terms cancel in pairs (Problem 2.1). So we obtain

$$\boxed{[J_i^L, J_j^L] = \mathrm{i}\varepsilon_{ijk}J_k^L, \qquad [J_i^R, J_j^R] = \mathrm{i}\varepsilon_{ijk}J_k^R, \qquad [J_i^L, J_j^R] = 0.} \tag{2.6}$$

Two completely independent angular momentum algebras! This is the trick promised at the start of the chapter.[2]

Now finding representations is easy. You learned in quantum mechanics that the finite-dimensional irreducible representations of one angular momentum algebra are labeled by $j = 0, \frac{1}{2}, 1, \frac{3}{2}, \ldots$, with dimension $2j + 1$. We have two independent copies of angular momentum, so the finite-dimensional irreducible representations of the Lorentz group are labeled by a pair of numbers (j_L, j_R), with dimension

$$\dim(j_L, j_R) = (2j_L + 1)(2j_R + 1). \tag{2.7}$$

The most common ones are:

[2] Strictly speaking, we used the complex coefficient i in the combinations (2.3), so the conclusion here holds "after complexification": $\mathfrak{so}(1,3) \otimes \mathbb{C} \cong \mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$. As a real Lie algebra, the Lorentz algebra is not two copies of $\mathfrak{su}(2)$, and the Lorentz group is certainly not $\mathrm{SU}(2) \times \mathrm{SU}(2)$: the latter is compact and the former is not. For the purpose of finding finite-dimensional representations, however, the complexified statement is all we need.

Representation	Name	Complex dimension
$(0,0)$	scalar	1
$(\frac{1}{2},0)$	left-handed Weyl spinor	2
$(0,\frac{1}{2})$	right-handed Weyl spinor	2
$(\frac{1}{2},\frac{1}{2})$	four-vector	4
$(\frac{1}{2},0)\oplus(0,\frac{1}{2})$	Dirac spinor (reducible)	4
$(1,0)\oplus(0,1)$	antisymmetric tensor	6

Ordinary rotations are generated by $R_i = J_i^L + J_i^R$, the sum of the two angular momenta. So under spatial rotations, (j_L, j_R) decomposes according to the rules of angular momentum addition into spins $|j_L - j_R|, \ldots, j_L + j_R$. For example, the four-vector $(\frac{1}{2},\frac{1}{2})$ splits into $j=0$ and $j=1$: the time component is a rotational scalar, and the three spatial components form a rotational vector. This is a reminder that you cannot say a field describes "four spin states" just because it has four components. In Chapter 7 we will see that the photon's four-component vector field actually describes only two physical states.

Example 2.1 The generators of the representation $(\frac{1}{2},0)$

Write down explicit matrices for R_i and K_i in the representation $(\frac{1}{2},0)$.

Solution In this representation J^R acts as zero, and J^L is the spin-$\frac{1}{2}$ angular momentum matrix $\sigma_i/2$ (the σ_i are the Pauli matrices, which will be introduced properly in the next section). Then

$$R_i = J_i^L + J_i^R = \frac{\sigma_i}{2}, \qquad K_i = -\mathrm{i}(J_i^L - J_i^R) = -\mathrm{i}\frac{\sigma_i}{2}.$$

Let's check one relation: $[K_1, K_2] = (-\mathrm{i})^2[\sigma_1/2, \sigma_2/2] = -\mathrm{i}\sigma_3/2 = -\mathrm{i}R_3$, in agreement with Eq. (2.2). Notice that K_i is anti-Hermitian, so the boost $e^{-\mathrm{i}\chi K_3} = e^{-\chi\sigma_3/2}$ is a real, nonunitary matrix. Boosts are not unitary in finite-dimensional representations; we will run into this again in Chapter 3. □

One last detail. If the generators R_i and K_i are all Hermitian operators (as they are in a unitary representation on a space of quantum states), then $(J_i^L)^\dagger = \frac{1}{2}(R_i - \mathrm{i}K_i) = J_i^R$: Hermitian conjugation swaps the two copies of angular momentum. In finite-dimensional representations K is not Hermitian, so this equation itself fails (look at the example above), but the fact behind it still holds: *taking the complex conjugate turns* (j_L, j_R) *into* (j_R, j_L). So the complex conjugate of a left-handed spinor is a right-handed spinor. We will need this very shortly.

Problem 2.1 Use Eq. (2.2) to verify that $[J_1^L, J_2^R] = 0$, and point out which terms cancel against which.

Problem 2.2 Compute $[L_{12}, L_{23}]$ from Eq. (1.33), and show that it is just $[R_3, R_1] = \mathrm{i}R_2$.

Problem 2.3 What is the dimension of the representation $(1,\frac{1}{2})$? Into which spins does it decompose under spatial rotations? Check that the dimensions add up.

2.3 Writing a Four-Vector as a Matrix

Next we are going to do something that looks like a magic trick: write a four-vector as a 2×2 matrix. This trick will lead us straight to spinors.

First, let me bring in the Pauli matrices:

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{2.8}$$

which satisfy $\sigma_i \sigma_j = \delta_{ij}\mathbf{1} + \mathrm{i}\varepsilon_{ijk}\sigma_k$, $\operatorname{tr}\sigma_i = 0$, $\operatorname{tr}(\sigma_i\sigma_j) = 2\delta_{ij}$. Adding the identity matrix $\mathbf{1}$, we define two sets of four matrices:

$$\sigma^\mu = (\mathbf{1}, \sigma_1, \sigma_2, \sigma_3), \qquad \bar{\sigma}^\mu = (\mathbf{1}, -\sigma_1, -\sigma_2, -\sigma_3). \tag{2.9}$$

Any 2×2 Hermitian matrix can be written as

$$H = \begin{pmatrix} a & u - \mathrm{i}v \\ u + \mathrm{i}v & d \end{pmatrix}, \qquad a, d, u, v \in \mathbb{R},$$

exactly four real numbers, the same as the number of components of a four-vector. That suggests we define

$$X \equiv x_\mu \sigma^\mu = \begin{pmatrix} x_0 + x_3 & x_1 - \mathrm{i}x_2 \\ x_1 + \mathrm{i}x_2 & x_0 - x_3 \end{pmatrix} = \begin{pmatrix} t - z & -x + \mathrm{i}y \\ -x - \mathrm{i}y & t + z \end{pmatrix}. \tag{2.10}$$

(The second equality uses $x_\mu = (t, -x, -y, -z)$.) Now compute its determinant:

$$\det X = (x_0 + x_3)(x_0 - x_3) - (x_1 - \mathrm{i}x_2)(x_1 + \mathrm{i}x_2) = x_0^2 - x_1^2 - x_2^2 - x_3^2 = x^\mu x_\mu. \tag{2.11}$$

This is the heart of the whole trick: the Minkowski length squared has turned into the determinant of a matrix.

How do we read x^μ back out of X? With a trace. From the properties of the Pauli matrices,

$$\operatorname{tr}(\sigma^\mu \bar{\sigma}^\nu) = 2\eta^{\mu\nu}. \tag{2.12}$$

($\mu = \nu = 0$ gives $\operatorname{tr}\mathbf{1} = 2$; $\mu = i, \nu = j$ gives $-\operatorname{tr}(\sigma_i\sigma_j) = -2\delta_{ij}$; one time and one space index give zero.) Therefore

$$\frac{1}{2}\operatorname{tr}(X\bar{\sigma}^\mu) = \frac{1}{2}x_\nu \operatorname{tr}(\sigma^\nu \bar{\sigma}^\mu) = x_\nu \eta^{\nu\mu} = x^\mu, \qquad \text{that is,} \qquad \boxed{x^\mu = \frac{1}{2}\operatorname{tr}(X\bar{\sigma}^\mu).} \tag{2.13}$$

Here $\bar{\sigma}^\mu$ plays the role of an "inverse in the sense of the trace". It is certainly not the ordinary inverse matrix of σ^μ: $\sigma_1^{-1} = \sigma_1$, while $\bar{\sigma}^1 = -\sigma_1$. While we're at it, let me record an identity that we will use again and again; it too can be checked directly from the multiplication rule of the Pauli matrices:

$$\sigma^\mu\bar{\sigma}^\nu + \sigma^\nu\bar{\sigma}^\mu = 2\eta^{\mu\nu}\mathbf{1}, \qquad \bar{\sigma}^\mu\sigma^\nu + \bar{\sigma}^\nu\sigma^\mu = 2\eta^{\mu\nu}\mathbf{1}. \tag{2.14}$$

Problem 2.4 For $x^\mu = (3, 1, 2, 0)$, write down the matrix X, compute $\det X$, and then use Eq. (2.13) to read back x^1.

2.4 $\mathrm{SL}(2,\mathbb{C})$

Now let X transform: take an arbitrary 2×2 complex matrix M and set

$$\boxed{X' = MXM^\dagger.} \tag{2.15}$$

This has two nice properties. First, X' is still Hermitian: $(MXM^\dagger)^\dagger = MX^\dagger M^\dagger = X'$, so it still corresponds to a real four-vector x'. Second, $\det X' = |\det M|^2 \det X$, so as long as $|\det M| = 1$ the interval is unchanged. An overall phase of M cancels out of $MXM^\dagger$ by itself ($e^{\mathrm{i}\theta}e^{-\mathrm{i}\theta} = 1$), so we may as well demand

$$\det M = 1. \tag{2.16}$$

The group of complex matrices satisfying this condition is called $\mathrm{SL}(2,\mathbb{C})$: the S stands for "special" (determinant 1), the L for "linear", and $\mathbb{C}$ says the matrix elements are complex. Count the parameters: a complex 2×2 matrix has 8 real parameters, and $\det M = 1$ is one complex equation, which removes two, leaving 6, the same number as the Lorentz group. This is no coincidence.

Because (2.15) is linear in X, it induces a linear transformation of x, and it preserves the interval, so it is a Lorentz transformation. Which one, exactly? Read it off using (2.13):

$$x'^\mu = \frac{1}{2}\operatorname{tr}(\bar\sigma^\mu X') = \frac{1}{2}\operatorname{tr}(\bar\sigma^\mu MXM^\dagger) = \frac{1}{2}\operatorname{tr}(\bar\sigma^\mu M\sigma_\nu M^\dagger)\, x^\nu, \tag{2.17}$$

(where we have written X as $x^\nu\sigma_\nu$). Comparing with $x'^\mu = \Lambda^\mu{}_\nu x^\nu$,

$$\boxed{\Lambda^\mu{}_\nu(M) = \frac{1}{2}\operatorname{tr}\left(\bar\sigma^\mu M\sigma_\nu M^\dagger\right).} \tag{2.18}$$

Taking $M = \mathbf{1}$, the relation $\operatorname{tr}(\bar\sigma^\mu\sigma_\nu) = 2\delta^\mu{}_\nu$ immediately gives $\Lambda = \mathbf{1}$; that is the most basic check.

Example 2.2 A diagonal matrix is a boost

Take $M_B(\chi) = \mathrm{diag}(e^{-\chi/2}, e^{\chi/2})$, and find the corresponding Lorentz transformation.

Solution $\det M_B = 1$, and M_B is real and diagonal, so $M_B^\dagger = M_B$. Multiply Eq. (2.10) by it once on the left and once on the right: the diagonal elements get multiplied by $e^{-\chi}$ and e^{χ} respectively, and the off-diagonal elements by $e^{-\chi/2}e^{\chi/2} = 1$. So

$$t' - z' = e^{-\chi}(t - z), \qquad t' + z' = e^{\chi}(t + z), \qquad x' = x, \qquad y' = y.$$

Solving, $t' = t\cosh\chi + z\sinh\chi$ and $z' = t\sinh\chi + z\cosh\chi$. That is exactly Eq. (1.14). A 2×2 matrix with just two diagonal entries has encoded a boost in four-dimensional spacetime. □

2.4.1 The Double Cover

You may already have noticed one thing: M and $-M$ give exactly the same transformation, because $(-M)X(-M)^\dagger = MXM^\dagger$. Are there any other M's that also correspond to the identity transformation? Suppose $MXM^\dagger = X$ for every Hermitian matrix X. Taking $X = \mathbf{1}$ gives $MM^\dagger = \mathbf{1}$; then taking $X = \sigma_1, \sigma_2, \sigma_3$ in turn gives $M\sigma_i M^{-1} = \sigma_i$, i.e., M commutes with all the Pauli matrices. A 2×2 matrix that commutes with all three Pauli matrices can only be a multiple of the identity, $c\mathbf{1}$, and then $\det M = c^2 = 1$ gives $c = \pm 1$. So there are exactly two: $\pm\mathbf{1}$.

On the other hand, SU(2) matrices implement all spatial rotations, and the example above implements a boost along z; combining the two you can get any Lorentz transformation. So every element of $\mathrm{SO}^+(1,3)$ corresponds to exactly two elements $\pm M$ of $\mathrm{SL}(2,\mathbb{C})$:

$$\boxed{\mathrm{SO}^+(1,3) \cong \mathrm{SL}(2,\mathbb{C})/\{\pm\mathbf{1}\}.} \tag{2.19}$$

We say that $\mathrm{SL}(2,\mathbb{C})$ is a **double cover** of the Lorentz group. Adding translations, the covering group of the Poincaré group is $\mathbb{R}^{1,3} \rtimes \mathrm{SL}(2,\mathbb{C})$, where $\mathrm{SL}(2,\mathbb{C})$ acts on the translation vector through $\Lambda(M)$.

2.5 Weyl Spinors

A double cover by itself might be no more than a mathematical curiosity, but it hints at something big in physics: since the matrices M of $\mathrm{SL}(2,\mathbb{C})$ are themselves 2×2 matrices, why not let them act directly on a two-component object?

$$\psi = \begin{pmatrix}\psi_1\\ \psi_2\end{pmatrix}, \qquad \psi_\alpha \to M_\alpha{}^\beta \psi_\beta. \tag{2.20}$$

An object that transforms this way is called a **Weyl spinor**. Its two components don't correspond to any "spatial directions"; they are simply the components along the two basis vectors of this representation. Its complex conjugate transforms with M^*, and to tell the two apart we put a dot on its index:

$$\bar\psi_{\dot\alpha} \equiv (\psi_\alpha)^* \to (M^*)_{\dot\alpha}{}^{\dot\beta}\bar\psi_{\dot\beta}. \tag{2.21}$$

The dot is purely a piece of notation; it has nothing whatever to do with a time derivative. Following the pattern of the previous section, these two kinds of spinor belong to $(\frac12,0)$ and $(0,\frac12)$ respectively:

$$\psi_\alpha \in \left(\tfrac12,0\right) \text{ (left-handed)}, \qquad \bar\psi_{\dot\alpha} \in \left(0,\tfrac12\right) \text{ (right-handed)}. \tag{2.22}$$

Verifying this requires the explicit relation between M and the generators, so we postpone it to Section 5.4, where we will compute it directly.

The most famous property of spinors is this. A rotation by angle θ about the z axis corresponds to

$$M_R(\theta) = e^{-\mathrm{i}\theta\sigma_3/2} = \begin{pmatrix} e^{-\mathrm{i}\theta/2} & 0\\ 0 & e^{\mathrm{i}\theta/2}\end{pmatrix} \tag{2.23}$$

(Problem 2.5 checks that it really does produce an ordinary rotation). Notice that $M_R(2\pi) = -\mathbf{1}$ and $M_R(4\pi) = \mathbf{1}$. After one full turn the four-vector is back where it started, but the spinor has changed sign; it takes two full turns to bring it back to itself. That sounds absurd, but there is no contradiction: the overall phase of an isolated quantum state is unobservable, but when the state is superposed with another state and they interfere, the minus sign shows up, and it has indeed been observed experimentally.

Finally, take one more look at $X_{\alpha\dot\alpha}$. It carries one undotted index and one dotted index; under $X \to MXM^\dagger$ the left one transforms with M and the right one with M^*. So the four-vector is the $(\frac12,\frac12)$ representation, consistent with the table in the previous section. This does not mean, though, that every four-vector is a product $\psi_\alpha\bar\chi_{\dot\alpha}$ of two spinors: such an outer product has rank at most 1, and a general 2×2 matrix needs a sum of several outer products.

***Problem 2.5** Compute $M_R(\theta) X M_R(\theta)^\dagger$ directly, and verify that it produces the usual rotation in the x, y plane. What happens when $\theta = 2\pi$?

****Problem 2.6** Prove that $\Lambda(M_1 M_2) = \Lambda(M_1)\Lambda(M_2)$. (Hint: don't grind through Eq. (2.18); just look at doing $X \mapsto MXM^\dagger$ twice in a row.)

Chapter 3

Particles: Mass, Spin and Helicity

Now we can give a serious answer to the question raised in the preface: what is "a kind of particle"? The answer Wigner gave in 1939 is astonishingly concise: *a kind of particle is an irreducible unitary representation of the Poincaré group.* The goal of this chapter is to understand every word of that sentence, and then to list all possible kinds of particle. You will see mass and spin both appear quite naturally as "labels of representations", with no extra assumptions needed.

3.1 Particle States and Field Components

First we need to clear up a point that is easy to muddle. The representations of the last chapter (scalars, spinors, four-vectors) describe how the components of a field *at a single spacetime point* mix, and they are finite-dimensional. The quantum state of a particle, say $|p, r\rangle$ (four-momentum p^μ, with r standing for the remaining "internal" labels), is a different matter altogether. Even if a particle has only one internal state, its momentum can take a continuum of values, so describing it takes infinitely many basis vectors. P^μ is an operator, and p^μ is its eigenvalue:[1]

$$P^\mu |p, r\rangle = p^\mu |p, r\rangle. \tag{3.1}$$

Symmetry transformations on quantum states must preserve probabilities, so they are implemented by unitary operators U, $U^\dagger U = \mathbf{1}$. But the example in the last chapter already told us that boosts are not unitary in finite-dimensional representations. This is no accident: the Lorentz group is noncompact (rapidity can be arbitrarily large), and one can prove mathematically that a noncompact simple Lie group has no finite-dimensional unitary representations apart from the trivial one. So the two divide the labor cleanly: finite-dimensional representations describe how the components of fields change, and they need not be unitary; quantum states need unitarity, so they can only live in infinite-dimensional spaces, whose infinitely many basis vectors are labeled precisely by the continuous momentum.

3.2 Casimir Operators

How do we stick a stable label on a kind of particle? Energy won't do: switch to another observer and the energy changes. Surely the same particle doesn't "become a different kind of particle" as a result? What we need are quantities that are invariant under all Poincaré transformations.

An operator that commutes with all the generators of a group is called a **Casimir operator**.

[1] Strictly speaking, momentum eigenstates cannot be normalized; they are "generalized eigenstates", and genuine physical states are wave packets built by superposing them. This is exactly the same as with plane waves in nonrelativistic quantum mechanics.

Why is that useful? Suppose C commutes with every $U(g)$, and $C|\psi\rangle = c|\psi\rangle$. Then

$$C\,U(g)|\psi\rangle = U(g)\,C|\psi\rangle = c\,U(g)|\psi\rangle,$$

so the transformed state still has the same eigenvalue c. In an irreducible representation, every state can be reached from every other by group actions, so C is simply a constant on the whole representation. That constant is exactly the label we are looking for. The $\boldsymbol{J}^2$ of quantum mechanics is the Casimir operator of the rotation group, and its value $j(j+1)$ labels the spin.

3.2.1 The First Casimir: Mass

The most obvious candidate is

$$C_1 = P^\mu P_\mu. \tag{3.2}$$

It commutes with P, because $[P_\mu, P_\nu] = 0$. It also commutes with L: using Eq. (1.29) and the product rule for commutators,

$$\begin{aligned}[L_{\mu\nu}, P^\rho P_\rho] &= [L_{\mu\nu}, P^\rho]P_\rho + P^\rho[L_{\mu\nu}, P_\rho] \\ &= \mathrm{i}(P_\mu P_\nu - P_\nu P_\mu) + \mathrm{i}(P_\nu P_\mu - P_\mu P_\nu) = 0.\end{aligned} \tag{3.3}$$

On the states of a particle of mass m,

$$\boxed{P^2|p,r\rangle = m^2|p,r\rangle,} \tag{3.4}$$

which is the operator version of $E^2 - \boldsymbol{p}^2 = m^2$.

3.2.2 The Second Casimir: the Pauli–Lubanski Vector

Yet particles with the same mass can still differ, for instance in spin. We want an invariant that describes spin. The natural first instinct is to use the squared angular momentum $\boldsymbol{R}^2$, but unfortunately it is not invariant under boosts, and it also has orbital angular momentum mixed in, which depends on the choice of origin. The way out is the following clever combination, called the **Pauli–Lubanski vector**:

$$\boxed{W^\mu = \frac{1}{2}\varepsilon^{\mu\nu\rho\sigma}P_\nu L_{\rho\sigma}.} \tag{3.5}$$

Here $\varepsilon^{\mu\nu\rho\sigma}$ is the totally antisymmetric symbol in four dimensions, and we take $\varepsilon^{0123} = -1$ (so that $\varepsilon_{0123} = +1$).[2]

W^μ has two basic properties:

$$W^\mu P_\mu = 0, \qquad [W^\mu, P^\nu] = 0. \tag{3.6}$$

The first you can see almost at a glance: $W^\mu P_\mu = \frac{1}{2}\varepsilon^{\mu\nu\rho\sigma}P_\mu P_\nu L_{\rho\sigma}$ (the commutator terms produced when moving P_μ past $L_{\rho\sigma}$ vanish for the same reason), and $P_\mu P_\nu$ is symmetric in $\mu\nu$ while ε is antisymmetric in $\mu\nu$, so contracting them gives zero. This "symmetric times antisymmetric equals zero" argument will come up again and again in this book; it's worth remembering. The second property relies on the same mechanism:

$$[W^\mu, P_\lambda] = \frac{1}{2}\varepsilon^{\mu\nu\rho\sigma}P_\nu[L_{\rho\sigma}, P_\lambda] = \frac{\mathrm{i}}{2}\varepsilon^{\mu\nu\rho\sigma}P_\nu\big(\eta_{\sigma\lambda}P_\rho - \eta_{\rho\lambda}P_\sigma\big) = 0. \tag{3.7}$$

[2] The advantage of this convention is that in the rest frame $\boldsymbol{W} = +m\boldsymbol{R}$; see Eq. (3.14) below. If your book takes $\varepsilon^{0123} = +1$, all of W^μ changes sign, but W^2 and every conclusion drawn from it stays the same.

Using the Poincaré algebra one can also prove (we won't go through it here)

$$[L_{\mu\nu}, W_\rho] = \mathrm{i}(\eta_{\nu\rho}W_\mu - \eta_{\mu\rho}W_\nu), \qquad [W^\mu, W^\nu] = -\mathrm{i}\varepsilon^{\mu\nu\rho\sigma}W_\rho P_\sigma. \tag{3.8}$$

The first says W is a four-vector, so its "length squared" is invariant under Lorentz transformations; combined with $[W, P] = 0$, this means

$$\boxed{C_2 = W^\mu W_\mu} \tag{3.9}$$

commutes with all the generators: it is the second Casimir operator. Notice that the individual components of W do not commute with one another; what is invariant is the combination W^2 built from them.

Why do we say that W measures only intrinsic spin? If you plug in just the orbital part $L_{\rho\sigma} = x_\rho P_\sigma - x_\sigma P_\rho$, you will find that W^μ vanishes identically: the symmetric product $P_\nu P_\sigma$ shows up inside the ε. So W automatically strips out the orbital angular momentum, and all that remains is the contribution of Σ, namely the spin.

3.3 Wigner's Little Group Method

With two candidate labels in hand, we still need a systematic way to actually construct the representations. Wigner's idea was this: first pick a particularly simple four-momentum k^μ, called the **standard momentum**, and study only the states with that one momentum; then use boosts to "push" them out to arbitrary momenta.

Concretely, let $B(p)$ be some Lorentz transformation that takes k into p, $p = B(p)k$. We define $|p, r\rangle \equiv U(B(p))|k, r\rangle$. Now apply an arbitrary Lorentz transformation Λ to $|p, r\rangle$; it will turn into some state with momentum Λp. The crux is the following combination:

$$w(\Lambda, p) = B(\Lambda p)^{-1}\,\Lambda\,B(p). \tag{3.10}$$

It first pushes k to p, then transforms to Λp, and finally pushes back to k:

$$w(\Lambda, p)\,k = B(\Lambda p)^{-1}\Lambda p = k.$$

So w belongs to the set of Lorentz transformations that leave k unchanged. The group these transformations form is called the **little group** of k:

$$G_k = \{\Lambda \in \mathrm{SO}^+(1,3) \mid \Lambda k = k\}. \tag{3.11}$$

The little group doesn't change the momentum, but it can change the internal state (the orientation of the spin, for example). The whole story then boils down to: *representation of the Poincaré group = "orbit" of momenta + representation of the little group.* To classify particles, all we have to do is classify the representations of the little group.

3.4 Massive Particles

If $m > 0$, the most natural standard momentum is the rest momentum

$$k^\mu = (m, 0, 0, 0). \tag{3.12}$$

Which transformations are in the little group? From $\Lambda k = k$ we get ${\Lambda^0}_0 = 1$ and ${\Lambda^i}_0 = 0$; the Lorentz condition (1.11) then forces ${\Lambda^0}_i = 0$, and the remaining spatial block r satisfies $r^{\mathsf{T}} r = \mathbf{1}$, $\det r = 1$. So

$$\Lambda = \begin{pmatrix} 1 & 0 \\ 0 & r \end{pmatrix}, \qquad r \in \mathrm{SO}(3). \tag{3.13}$$

Physically this could not be more obvious: for a particle at rest, you can rotate the coordinates however you like without changing its momentum, but you cannot boost it, or it will start moving. In the covering group $\mathrm{SL}(2,\mathbb{C})$ the corresponding little group is SU(2), so half-integer spins are allowed too.

Let's compute W^μ in the rest frame. $P_\nu = (m,0,0,0)$ has only the $\nu = 0$ term, so $W^0 = \frac{1}{2}\varepsilon^{00\rho\sigma} m L_{\rho\sigma} = 0$, while

$$W^0 = 0, \qquad W^i = \frac{1}{2}\varepsilon^{i0jk}\, m\, L_{jk} = \frac{m}{2}\varepsilon_{ijk} L_{jk} = m R_i. \tag{3.14}$$

(In the middle we used $\varepsilon^{i0jk} = -\varepsilon^{0ijk} = \varepsilon_{ijk}$; this is where $\varepsilon^{0123} = -1$ comes in.) The R_i here act on the internal states of the particle at rest; they are its spin operators. If the spin quantum number is s, i.e., $\boldsymbol{R}^2 = s(s+1)$, then

$$W^2 = (W^0)^2 - \sum_i (W^i)^2 = -m^2 \boldsymbol{R}^2, \tag{3.15}$$

that is,

$$\boxed{C_1 = m^2, \qquad C_2 = -m^2 s(s+1).} \tag{3.16}$$

The minus sign comes from the metric, and $s(s+1)$ from the angular momentum algebra. Although we computed in the rest frame, C_1 and C_2 are invariants, so this result holds for all the momentum states of the same particle.

How many internal states are there at fixed momentum? This is a calculation you did in quantum mechanics: use the ladder operators $R_\pm = R_1 \pm \mathrm{i} R_2$; from $[R_3, R_\pm] = \pm R_\pm$ you know that $R_\pm$ raise or lower the eigenvalue of R_3 by 1; a finite-dimensional representation requires the ladder to be capped at both ends, so s can only be $0, \frac{1}{2}, 1, \ldots$, and R_3 takes the values $-s, -s+1, \ldots, s$, $2s+1$ values in all. For $s = 0$ there is only one state, and $C_2 = 0$; for $s = \frac{1}{2}$ there are two states, and $C_2 = -\frac{3}{4}m^2$; for $s = 1$ there are three states, and $C_2 = -2m^2$.

Problem 3.1 For a particle of mass m and spin 1, write down the values of C_1 and C_2 and the number of internal states at fixed momentum. Why is the full space of one-particle states not three-dimensional?

3.5 Massless Particles

Massless particles are a bit more troublesome, because they have no rest frame. For a lightlike momentum with positive energy, $p^2 = 0$ and $E = |\boldsymbol{p}| > 0$. If a rest frame existed, we would have $\boldsymbol{p} = 0$ there, hence $E = 0$, and the whole four-momentum would vanish; but an invertible Lorentz transformation cannot turn a nonzero vector into zero. You can never catch up with a beam of light.

As the next best thing, we first rotate the coordinates so that the momentum lies along the z axis, and then take

$$k^\mu = (\kappa, 0, 0, \kappa), \qquad \kappa > 0. \tag{3.17}$$

κ is just a number picked to write down the standard momentum; it is not an intrinsic quantity of any kind: boost along z and κ gets multiplied by a factor e^χ.

3.5.1 The Little Group

Rotations about the z axis obviously leave k unchanged. Is there anything else? Yes, and somewhat unexpectedly: certain combinations of rotations and boosts work too. To find them, first read off from Eq. (1.29) how rotations and boosts change the components of the four-momentum:

$$[R_i, P^j] = \mathrm{i}\varepsilon_{ijk}P^k, \qquad [K_i, P^0] = -\mathrm{i}P^i, \qquad [K_i, P^j] = -\mathrm{i}\delta_{ij}P^0. \tag{3.18}$$

(Take the second as an example: $[K_i, P_0] = -[L_{0i}, P_0] = -\mathrm{i}(\eta_{i0}P_0 - \eta_{00}P_i) = \mathrm{i}P_i = -\mathrm{i}P^i$.) For a generator G to leave k unchanged means that all the $[G, P^\mu]$ vanish at $P = k$. Consider

$$N_1 = R_1 + K_2, \qquad N_2 = R_2 - K_1, \qquad H = R_3. \tag{3.19}$$

Let's check N_1 term by term:

$$\begin{aligned} &[N_1, P^0] = [K_2, P^0] = -\mathrm{i}P^2, && [N_1, P^1] = 0, \\ &[N_1, P^2] = [R_1, P^2] + [K_2, P^2] = \mathrm{i}(P^3 - P^0), && [N_1, P^3] = [R_1, P^3] = -\mathrm{i}P^2. \end{aligned} \tag{3.20}$$

At $k = (\kappa, 0, 0, \kappa)$ we have $P^2 = 0$ and $P^3 - P^0 = 0$, so all four vanish. Intuitively: rotate a little about the x axis, and a particle moving along z picks up a bit of momentum in the y direction; then boost a little along y, which cancels it exactly. The check for N_2 is entirely the same (Problem 3.2).

These three generators satisfy

$$\boxed{[N_1, N_2] = 0, \qquad [H, N_1] = \mathrm{i}N_2, \qquad [H, N_2] = -\mathrm{i}N_1.} \tag{3.21}$$

If you think of N_1, N_2 as two translations in a plane and of H as a rotation in that plane, this is exactly the algebra of the Euclidean group of the plane: the two translations commute with each other, and the rotation turns the direction of the translations. So the little group of a massless particle is isomorphic to

$$\mathrm{ISO}(2) = \mathbb{R}^2 \rtimes \mathrm{SO}(2). \tag{3.22}$$

("Translation" here is only an abstract way of speaking; N_1, N_2 are actually combinations of rotations and boosts, and have nothing to do with translations in spacetime.)

3.5.2 Helicity

N_1 and N_2 commute with each other and are both Hermitian, so they can be diagonalized simultaneously, and their eigenvalues form a two-dimensional vector (n_1, n_2). If some state had $(n_1, n_2) \neq 0$, then rotating it with H would produce a whole circle of different eigenvalues, and there would be infinitely many internal states at fixed momentum: a so-called "continuous spin" representation. No such particle has ever been seen in nature, so we consider only

$$N_1|\mathrm{phys}\rangle = N_2|\mathrm{phys}\rangle = 0. \tag{3.23}$$

That leaves, as the only part of the little group that really does anything, the SO(2) generated by $H = R_3$. SO(2) is an abelian group, and its irreducible representations are all one-dimensional: just a phase,

$$e^{-\mathrm{i}\theta H}|k,h\rangle = e^{-\mathrm{i}\theta h}|k,h\rangle. \tag{3.24}$$

h is called the **helicity**. The standard momentum points along z, so h is the projection of the angular momentum onto the direction of motion. For a momentum in a general direction,

$$h = \frac{\boldsymbol{R}\cdot\boldsymbol{p}}{|\boldsymbol{p}|}. \tag{3.25}$$

The orbital angular momentum $\boldsymbol{x}\times\boldsymbol{p}$ is perpendicular to $\boldsymbol{p}$, so what is measured here is only the intrinsic angular momentum. Since spinors can change sign under a 2π rotation, h can be an integer or a half-integer.

3.5.3 Both Casimirs Vanish

On a state of fixed momentum $p^\mu = (E,\boldsymbol{p})$, replace P_ν by the numbers $p_\nu = (E,-\boldsymbol{p})$ and expand Eq. (3.5):

$$W^0 = \boldsymbol{p}\cdot\boldsymbol{R}, \qquad \boldsymbol{W} = E\boldsymbol{R} - \boldsymbol{p}\times\boldsymbol{K}. \tag{3.26}$$

(Derivation of W^0: $W^0 = \frac{1}{2}\varepsilon^{0jkl}p_jL_{kl}$, with $\varepsilon^{0jkl} = -\varepsilon_{jkl}$ and $p_j = -p^j$; the two minus signs cancel, giving $\frac{1}{2}\varepsilon_{jkl}p^jL_{kl} = \boldsymbol{p}\cdot\boldsymbol{R}$. The derivation for $\boldsymbol{W}$ is similar, and I leave it to you.) At rest this reduces to Eq. (3.14). At $\boldsymbol{p} = (0,0,\kappa)$, $E=\kappa$, we have $\boldsymbol{p}\times\boldsymbol{K} = \kappa(-K_2,K_1,0)$, so

$$W^\mu = \kappa\,(R_3,\ R_1+K_2,\ R_2-K_1,\ R_3) = \kappa\,(H,N_1,N_2,H). \tag{3.27}$$

All three generators of the little group show up in W^μ. That's no surprise: W^μ is, by its very nature, the covariant way of writing "the part of the Lorentz transformations that leaves the momentum unchanged". In the representation with $N_1 = N_2 = 0$ and $H = h$,

$$\boxed{W^\mu = hP^\mu, \qquad P^2 = 0, \qquad W^2 = h^2P^2 = 0.} \tag{3.28}$$

Both Casimirs vanish! That does not mean a massless particle carries no spin information. A lightlike vector can be nonzero and still have zero length squared, and W^μ is precisely such a nonzero vector, proportional to P^μ. The real label is the proportionality constant h, which has to be specified separately.

The photon, for example, has two helicities, $h=+1$ and $h=-1$, and no $h=0$ state. For the connected Poincaré group, a single fixed h is already an irreducible representation by itself; the reason both ±1 appear for the photon is that the CPT transformation turns a particle of helicity h into an antiparticle of helicity $-h$, and the photon is its own antiparticle. In Chapter 5 we will see another example: a massless Weyl field gives a particle with $h=-\frac{1}{2}$, whose antiparticle has $h=+\frac{1}{2}$.

Let's put the two cases side by side:

	Massive, $m>0$	Massless
Standard momentum	$(m,0,0,0)$	$(\kappa,0,0,\kappa)$
Little group	SO(3), covered by SU(2)	ISO(2), translations act as zero
Labels	mass m, spin s	helicity h
Internal states at fixed momentum	$2s+1$	one for each h
$C_1,\ C_2$	$m^2,\ -m^2s(s+1)$	$0,\ 0$

***Problem 3.2** Verify that $N_2 = R_2 - K_1$ leaves the momentum unchanged at $k = (\kappa, 0, 0, \kappa)$.

Problem 3.3 For a massless particle, show that $W^\mu = hP^\mu$ implies $W^2 = 0$. Can you conclude from $W^2 = 0$ that $h = 0$?

Problem 3.4 A massive particle moves along the z direction, $p^\mu = (E, 0, 0, p)$. Use Eq. (3.26) to show that $W^0/|\boldsymbol{p}|$ is its helicity.

Chapter 4

Scalar Fields

The classification in the last chapter was carried out at the level of quantum states. But when we write down a theory, we don't write states directly; we write *fields* and their Lagrangians, and particles appear as excitations of the fields only after quantization. So what the next few chapters do is turn each representation into a kind of field, write down its Lagrangian and equation of motion, and then check: are the mass and spin given by the equation exactly the labels of Chapter 3?

We start with the simplest case: the scalar field, which has no components at all to rotate.

4.1 How a Scalar Field Transforms

A field is a rule that assigns a quantity to every spacetime point. A real scalar field $\phi(x)$ gives a real number at each point, and a complex scalar field $\varphi(x)$ gives a complex number. "Real" and "complex" refer to the values the field takes; "scalar" refers to how it behaves under Lorentz transformations. These are two different things.

The transformation law for a scalar field is

$$\boxed{\phi'(x') = \phi(x), \qquad x' = \Lambda x + a.} \tag{4.1}$$

What this means is: the same physical event is called x' in the new coordinates and x in the old ones, and the two observers measure the same value of the field there. If you want to write both fields with the same argument, it reads

$$\phi'(x) = \phi\big(\Lambda^{-1}(x - a)\big). \tag{4.2}$$

So "a scalar field is invariant" does not mean $\phi'(x) = \phi(x)$. The argument of the field still has to change; it's just that there are no components to mix. The simplest example is a pure translation $x' = x + a$: then $\phi'(x) = \phi(x - a)$. If the original field has a peak at the origin, the new field has a peak at a. That is exactly "shifting the whole field profile by a".

4.2 The Lagrangian

In field theory what we write down is a *Lagrangian density* $\mathcal{L}$; integrating it over space gives the Lagrangian, and integrating that over time gives the action:

$$L(t) = \int \mathrm{d}^3x\, \mathcal{L}, \qquad S[\phi] = \int \mathrm{d}^4x\, \mathcal{L}. \tag{4.3}$$

(Physicists are usually lazy and call $\mathcal{L}$ the Lagrangian too; so does this book.) The square brackets remind you that S depends on the entire function $\phi(x)$, not on a single number.

We want a free theory that is local, Lorentz invariant, and has equations of motion of at most second order. The simplest choice is

$$\boxed{\mathcal{L} = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \frac{1}{2}m^2\phi^2.} \tag{4.4}$$

Spelling out the indices,

$$\mathcal{L} = \frac{1}{2}(\partial_t\phi)^2 - \frac{1}{2}(\nabla\phi)^2 - \frac{1}{2}m^2\phi^2. \tag{4.5}$$

The first term is the "kinetic energy", the second the gradient energy, and the third the mass term. The $\frac{1}{2}$ in front is just a conventional normalization; it will cancel against the 2 produced by differentiation when we vary the action in a moment. The time-derivative term must be positive, or the energy would be unbounded below; with our metric, $\partial_\mu\phi\,\partial^\mu\phi$ happens to give the time derivative a plus sign and the space derivatives a minus sign, so overall we take $+\frac{1}{2}$. If we wanted to add interactions, we could add terms such as $-\lambda\phi^4/4!$; in this chapter we study only the free field.

Why is (4.4) Lorentz invariant? Applying the chain rule to (4.1), $\partial'_\mu\phi'(x') = (\Lambda^{-1})^\nu{}_\mu\,\partial_\nu\phi(x)$; when two such factors are contracted with the metric, the inverse matrices are swallowed up by the Lorentz condition, so $\partial'_\mu\phi'\,\partial'^\mu\phi' = \partial_\mu\phi\,\partial^\mu\phi$. ϕ^2 is a scalar to begin with, and the volume element $\mathrm{d}^4x' = |\det\Lambda|\,\mathrm{d}^4x = \mathrm{d}^4x$ is unchanged too, so the action is invariant.

4.2.1 Dimensional Analysis

In natural units every quantity can be measured as a power of mass. We write $[\,\cdot\,]$ for the mass dimension of a quantity: for example, $[m] = 1$, $[x] = -1$, $[\partial_\mu] = 1$. The action appears in the exponent of $e^{\mathrm{i}S/\hbar}$, so it must be dimensionless; $[\mathrm{d}^4x] = -4$, so $[\mathcal{L}] = 4$. The kinetic term contains two ϕ's and two derivatives, so

$$2[\phi] + 2 = 4 \qquad\Longrightarrow\qquad [\phi] = 1. \tag{4.6}$$

In a general D-dimensional spacetime the same argument gives $[\phi] = (D-2)/2$. This kind of simple counting is surprisingly useful in field theory; in Chapter 5 we will use it again to find the dimension of a spinor field.

4.3 The Equation of Motion

Give the field an arbitrary small change $\phi \to \phi + \delta\phi$, and require $\delta\phi = 0$ on the boundary. The first-order change in the action is

$$\begin{aligned}\delta S &= \int \mathrm{d}^4x \left[\frac{\partial\mathcal{L}}{\partial\phi}\delta\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\,\partial_\mu\delta\phi\right] \\ &= \int \mathrm{d}^4x \left[\frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right]\delta\phi + \int \mathrm{d}^4x\,\partial_\mu\left[\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\delta\phi\right].\end{aligned} \tag{4.7}$$

The first line uses $\delta(\partial_\mu\phi) = \partial_\mu(\delta\phi)$, and the second is an integration by parts. The final total-derivative integral turns into an integral over the boundary, where $\delta\phi = 0$, so it vanishes. $\delta\phi$ can be chosen arbitrarily in the interior, so for $\delta S = 0$ the bracket must vanish everywhere:

$$\boxed{\partial_\mu\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} - \frac{\partial\mathcal{L}}{\partial\phi} = 0.} \tag{4.8}$$

This is the Euler–Lagrange equation of field theory. As in classical mechanics, it does not require $\mathcal{L}$ to be a minimum at each point; it requires the whole action to be stationary under all small variations.

For Eq. (4.4), $\partial\mathcal{L}/\partial\phi = -m^2\phi$; if you don't trust yourself with the other derivative, write out the metric and then differentiate:

$$\frac{\partial}{\partial(\partial_\mu\phi)}\Big(\frac{1}{2}\eta^{\rho\sigma}\partial_\rho\phi\,\partial_\sigma\phi\Big) = \frac{1}{2}\big(\eta^{\mu\sigma}\partial_\sigma\phi + \eta^{\rho\mu}\partial_\rho\phi\big) = \partial^\mu\phi.$$

Plugging into (4.8),

$$\boxed{(\Box + m^2)\phi = 0, \qquad \Box \equiv \partial_\mu\partial^\mu = \partial_t^2 - \nabla^2.} \tag{4.9}$$

This is the **Klein–Gordon equation**. $\Box$ is called the d'Alembertian; it contains time derivatives, so don't confuse it with the Laplacian ∇^2.

4.4 Plane Waves and the Mass Shell

Try a plane wave,

$$\phi(x) = Ae^{-\mathrm{i}p\cdot x} = Ae^{-\mathrm{i}Et + \mathrm{i}\boldsymbol{p}\cdot\boldsymbol{x}}. \tag{4.10}$$

(A genuine solution for a real field is a combination of this and its complex conjugate; we use the complex exponential just for convenience.) Differentiating gives $\partial_\mu\phi = -\mathrm{i}p_\mu\phi$, so $\Box\phi = -p^2\phi$, and the Klein–Gordon equation becomes

$$(-p^2 + m^2)\phi = 0 \quad\Longrightarrow\quad p^2 = m^2 \quad\Longrightarrow\quad E^2 = \boldsymbol{p}^2 + m^2. \tag{4.11}$$

The parameter m in the Lagrangian really is the mass. Momenta satisfying $p^2 = m^2$ are said to be "on shell", because in $(E, \boldsymbol{p})$ space they lie on a hyperboloid. Expanding a general field configuration as a Fourier integral,

$$\phi(x) = \int \frac{\mathrm{d}^4p}{(2\pi)^4} e^{-\mathrm{i}p\cdot x}\tilde{\phi}(p), \tag{4.12}$$

the equation of motion becomes

$$\boxed{(p^2 - m^2)\,\tilde{\phi}(p) = 0.} \tag{4.13}$$

What it says is that the Fourier components of a free solution can be nonzero only on the mass shell.

Now let's connect this with Chapter 3. From $P_\mu = \mathrm{i}\partial_\mu$ we get $P^2 = -\Box$, so the Klein–Gordon equation is exactly

$$(P^2 - m^2)\phi = 0,$$

with the first Casimir operator taking the value m^2. A scalar field has no components, $\Sigma = 0$, so $W^\mu = 0$, and it describes a particle of spin zero: for $m > 0$ it is $(m, s = 0)$, and for $m = 0$ it is helicity $h = 0$. The equation of motion is the Casimir condition, and this pattern will repeat itself for every kind of field that follows.[1]

[1] Strictly speaking, "particles" only appear after the field is quantized; here we are just seeing that the free field equation matches the labels of the corresponding one-particle representation. After quantization, a free real scalar field describes a neutral spin-zero particle that is its own antiparticle.

Problem 4.1 Find the equation of motion for $\mathcal{L} = \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 - \lambda\phi^4/4!$. Why can the solutions no longer be regarded as a collection of independent plane waves?

***Problem 4.2** In D-dimensional spacetime, what is the mass dimension of the coupling constant g_n of an interaction term $g_n\phi^n$? For $D = 4$, what are the dimensions of the coupling constants of ϕ^3 and ϕ^4?

4.5 The Complex Scalar Field

Combine two real fields of the same mass into one complex field:

$$\boxed{\varphi = \frac{\phi_1 + \mathrm{i}\phi_2}{\sqrt{2}}, \qquad \varphi^* = \frac{\phi_1 - \mathrm{i}\phi_2}{\sqrt{2}}.} \tag{4.14}$$

Its Lagrangian is

$$\boxed{\mathcal{L} = \partial_\mu\varphi^*\,\partial^\mu\varphi - m^2\varphi^*\varphi.} \tag{4.15}$$

Notice there is no $\frac{1}{2}$ in front. Substitute (4.14) and you'll see why: $\varphi^*\varphi = \frac{1}{2}(\phi_1^2 + \phi_2^2)$, and the cross terms in the kinetic term cancel, one positive and one negative, leaving $\frac{1}{2}(\partial\phi_1)^2 + \frac{1}{2}(\partial\phi_2)^2$ (Problem 4.3). So (4.15) is exactly the sum of two canonically normalized real scalar fields, and the $\sqrt{2}$ is there precisely for that purpose.

To find the equations of motion, you could vary ϕ_1 and ϕ_2 separately; or you can use an equivalent but more economical method: treat φ and φ^* as two independent variables. This does not magically turn two real degrees of freedom into four; it just recombines the "real-part equation" and the "imaginary-part equation" in a different way. Varying with respect to φ^*, $\partial\mathcal{L}/\partial\varphi^* = -m^2\varphi$ and $\partial\mathcal{L}/\partial(\partial_\mu\varphi^*) = \partial^\mu\varphi$, so

$$(\Box + m^2)\varphi = 0, \tag{4.16}$$

while varying with respect to φ gives its complex conjugate.

Why bother introducing a complex field? Because it makes a symmetry obvious at a glance: (4.15) is invariant under

$$\varphi \to e^{\mathrm{i}\alpha}\varphi, \tag{4.17}$$

where α is a constant independent of the spacetime point. In the language of ϕ_1, ϕ_2 this just rotates the two real fields by an angle in their plane, but in the language of the complex field it is multiplication by a phase. This $U(1)$ symmetry corresponds to a conserved charge (electric charge, say), and after quantization the complex scalar field describes a particle and an antiparticle of opposite charges, both with spin zero. In supersymmetry, the partner of a Weyl spinor is precisely a complex scalar field.

That completes the story of the scalar field. It used only the most trivial representation, $(0, 0)$. The next three chapters apply the same procedure (write down the transformation law, write down an invariant Lagrangian, read off the Casimirs from the equation of motion) to fields with components: first spinors, then vectors.

Problem 4.3 Without relying on the conclusions in the text, expand $\partial_\mu \varphi^* \partial^\mu \varphi$ directly and show that the complex-field Lagrangian equals the sum of the Lagrangians of two real fields. What would happen if the definition had no $1/\sqrt{2}$?

Chapter 5

Spinor Fields

The scalar field is too simple: it has no components to rotate at all. In this chapter we turn to the really interesting cases: fields whose components transform as $(\frac{1}{2}, 0)$ and $(0, \frac{1}{2})$, that is, the fields that correspond to fermions such as electrons, neutrinos and quarks. Before writing down Lagrangians, we have to master one basic skill: how to "glue" spinors together into Lorentz invariants.

5.1 Spinor Indices

For four-vectors you are already well practiced at this skill: raise and lower indices with $\eta_{\mu\nu}$, contract one upper index with one lower index, and the result is a Lorentz invariant, for example $x \cdot y = x^\mu y_\mu$. Spinors need the same kind of tools. Recall from Chapter 2 that

$$\psi_\alpha \to M_\alpha{}^\beta \psi_\beta, \qquad \bar{\psi}_{\dot{\alpha}} \to (M^*)_{\dot{\alpha}}{}^{\dot{\beta}} \bar{\psi}_{\dot{\beta}}, \qquad M \in \mathrm{SL}(2, \mathbb{C}). \tag{5.1}$$

How do we build, out of two spinors ψ, χ, a number that doesn't change with M? Writing $\psi_1\chi_1 + \psi_2\chi_2$ won't work, because M is in general not an orthogonal matrix. We need a "metric" that is invariant under $\mathrm{SL}(2, \mathbb{C})$.

5.1.1 The Spinor Metric

The answer is the antisymmetric symbol. Define

$$\boxed{\varepsilon^{12} = -\varepsilon^{21} = 1, \qquad \varepsilon_{12} = -\varepsilon_{21} = -1,} \tag{5.2}$$

with all other components zero; the dotted versions $\varepsilon^{\dot{\alpha}\dot{\beta}}$, $\varepsilon_{\dot{\alpha}\dot{\beta}}$ take the same numerical values. As matrices,

$$\varepsilon^{\alpha\beta} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad \varepsilon_{\alpha\beta} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \varepsilon^{\alpha\beta}\varepsilon_{\beta\gamma} = \delta^\alpha{}_\gamma. \tag{5.3}$$

The upper and lower ε's differ by a minus sign precisely so that they are inverses of each other.

Why is ε invariant? Because for *any* 2×2 matrix M,

$$\boxed{M_\alpha{}^\gamma M_\beta{}^\delta \, \varepsilon_{\gamma\delta} = \det M \; \varepsilon_{\alpha\beta}.} \tag{5.4}$$

The proof takes just two lines. The left side is antisymmetric in α, β (swap $\alpha \leftrightarrow \beta$, rename the dummy indices γ, δ, and all that's left is a sign change of ε), and in two dimensions an antisymmetric object has only one independent component, so the left side must be proportional to $\varepsilon_{\alpha\beta}$. The constant of proportionality follows from a quick calculation with $\alpha = 1, \beta = 2$:

$$M_1{}^1 M_2{}^2 \varepsilon_{12} + M_1{}^2 M_2{}^1 \varepsilon_{21} = \varepsilon_{12}\big(M_1{}^1 M_2{}^2 - M_1{}^2 M_2{}^1\big) = \varepsilon_{12} \det M.$$

For $\mathrm{SL}(2, \mathbb{C})$, $\det M = 1$, so ε doesn't budge under the transformation. This is exactly like the definition of a Lorentz transformation as "η is unchanged". $\mathrm{SL}(2, \mathbb{C})$ is to ε what the Lorentz group is to η.

5.1.2 Raising, Lowering and Contracting

So we can use ε to raise and lower spinor indices:

$$\boxed{\psi^\alpha = \varepsilon^{\alpha\beta}\psi_\beta, \qquad \psi_\alpha = \varepsilon_{\alpha\beta}\psi^\beta, \qquad \bar\psi^{\dot\alpha} = \varepsilon^{\dot\alpha\dot\beta}\bar\psi_{\dot\beta}, \qquad \bar\psi_{\dot\alpha} = \varepsilon_{\dot\alpha\dot\beta}\bar\psi^{\dot\beta}.} \tag{5.5}$$

There is just one rule: ε is written on the left, and its *second* index is contracted with the spinor. In components, $\psi^1 = \psi_2$ and $\psi^2 = -\psi_1$. Let's lower it back as a check: $\psi_1 = \varepsilon_{12}\psi^2 = (-1)(-\psi_1) = \psi_1$. No problem.

How does a spinor with an upper index transform? Writing Eq. (5.4) as the matrix equation $M\varepsilon M^{\mathsf T} = \varepsilon$ (where this ε is the one with lower indices), we can rewrite it as $\varepsilon^{-1}M\varepsilon = (M^{\mathsf T})^{-1}$, so

$$\psi'^\alpha = \varepsilon^{\alpha\beta}M_\beta{}^\gamma\varepsilon_{\gamma\delta}\psi^\delta = \psi^\delta(M^{-1})_\delta{}^\alpha. \tag{5.6}$$

Upper indices transform with the inverse matrix. This is entirely analogous to the upper and lower indices of a four-vector, which transform with Λ and Λ^{-1} respectively, and it tells us directly how to build invariants: contract an upper index with a lower index, and M and M^{-1} cancel. Our convention is that undotted indices go "upper on the left, lower on the right", and dotted indices "lower on the left, upper on the right":

$$\boxed{\psi\chi \equiv \psi^\alpha\chi_\alpha, \qquad \bar\psi\bar\chi \equiv \bar\psi_{\dot\alpha}\bar\chi^{\dot\alpha}.} \tag{5.7}$$

Let's check: $\psi'^\alpha\chi'_\alpha = \psi^\delta(M^{-1})_\delta{}^\alpha M_\alpha{}^\gamma\chi_\gamma = \psi^\gamma\chi_\gamma$. In components,

$$\psi\chi = \psi^1\chi_1 + \psi^2\chi_2 = \psi_2\chi_1 - \psi_1\chi_2. \tag{5.8}$$

There is a trap here that almost everyone falls into at least once. ε is just a set of ordinary numbers and can be moved around freely in a product, but it is antisymmetric. So

$$\psi^\alpha\chi_\alpha = \varepsilon^{\alpha\beta}\psi_\beta\chi_\alpha = -\psi_\beta\,\varepsilon^{\beta\alpha}\chi_\alpha = -\psi_\beta\chi^\beta. \tag{5.9}$$

Swap the up and down positions, and you pick up a minus sign! For four-vectors $x^\mu y_\mu = x_\mu y^\mu$, but for spinors it is not so. That is why the order of contraction must be written strictly according to the convention (5.7).

5.1.3 Indices of the σ Matrices

Go back to $X_{\alpha\dot\alpha} = x_\mu(\sigma^\mu)_{\alpha\dot\alpha}$ from Chapter 2. It tells us that σ^μ naturally carries one undotted lower index and one dotted lower index. What do we get if we raise both indices?

$$(\bar\sigma^\mu)^{\dot\alpha\alpha} = \varepsilon^{\dot\alpha\dot\beta}\varepsilon^{\alpha\beta}(\sigma^\mu)_{\beta\dot\beta}. \tag{5.10}$$

I've already written the answer on the left in advance; let's verify it. Read as matrices (row index $\dot\alpha$, column index α), the right side is $\varepsilon\,(\sigma^\mu)^{\mathsf T}\varepsilon^{\mathsf T}$. Take $\mu = 1$ as an example:

$$\begin{pmatrix}0&1\\-1&0\end{pmatrix}\begin{pmatrix}0&1\\1&0\end{pmatrix}\begin{pmatrix}0&-1\\1&0\end{pmatrix} = \begin{pmatrix}0&-1\\-1&0\end{pmatrix} = -\sigma_1 = \bar\sigma^1.$$

$\mu = 0, 2, 3$ likewise give $\mathbf{1}, -\sigma_2, -\sigma_3$. So the $\bar\sigma^\mu$ of Chapter 2, that "inverse in the sense of the trace", is no new invention at all: it is simply σ^μ with its indices up.

The σ matrices have one more important property. Eq. (2.18) can be read backwards: any Hermitian matrix can be expanded as $H = \frac{1}{2}\,\mathrm{tr}(H\bar{\sigma}^\mu)\sigma_\mu$, and taking $H = M\sigma_\nu M^\dagger$, the expansion coefficients are exactly $\Lambda^\mu{}_\nu(M)$. Therefore

$$\boxed{M\sigma_\nu M^\dagger = \Lambda^\mu{}_\nu\,\sigma_\mu, \qquad M^\dagger\bar{\sigma}^\mu M = \Lambda^\mu{}_\nu\,\bar{\sigma}^\nu.} \tag{5.11}$$

There is a little trick to proving the second equation; see Problem 5.2. What these two equations say is that transforming the two spinor indices of σ with M and $M^\dagger$ has the same effect as transforming its vector index with Λ. We say that σ is an **invariant tensor**. The immediate consequence: contract away all the spinor indices, leaving only a single μ, and the resulting combination (for example $\bar{\psi}\bar{\sigma}^\mu\psi$) is a four-vector.

Problem 5.1 Let $\psi_\alpha = (a, b)$ and $\chi_\alpha = (c, d)$, treating the components as ordinary numbers for now. Write down ψ^α, χ^α, compute $\psi^\alpha\chi_\alpha$ and $\psi_\alpha\chi^\alpha$, and verify that they differ by a minus sign.

****Problem 5.2** (a) For an arbitrary 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, verify that $\varepsilon A^{\mathsf{T}}\varepsilon^{\mathsf{T}} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \equiv \mathrm{adj}\,A$ (the adjugate matrix). It follows that $\bar{\sigma}^\mu = \mathrm{adj}\,\sigma^\mu$. (b) Using $\mathrm{adj}(ABC) = \mathrm{adj}\,C\;\mathrm{adj}\,B\;\mathrm{adj}\,A$, and the fact that $\mathrm{adj}\,M = M^{-1}$ when $\det M = 1$, start from $X' = MXM^\dagger$ and prove the second of Eqs. (5.11).

5.2 The Dirac Equation

In 1928 Dirac was quite unhappy with the Klein–Gordon equation. It is second order in time, so the "probability density" derived from it is not positive definite. He wanted a relativistic equation that was first order in time, while still making every solution satisfy $p^2 = M^2$. The most direct attempt is to write an equation linear in the momentum:

$$(\gamma^\mu P_\mu - M)\Psi = 0, \qquad P_\mu = \mathrm{i}\partial_\mu, \tag{5.12}$$

where the γ^μ are four constant matrices yet to be determined, and Ψ is a column of components. Now we demand that "squaring" it brings us back to the mass-shell condition. Multiply on the left by $(\gamma^\nu P_\nu + M)$:

$$(\gamma^\nu P_\nu + M)(\gamma^\mu P_\mu - M)\Psi = \big(\gamma^\nu\gamma^\mu P_\nu P_\mu - M^2\big)\Psi = \Big(\tfrac{1}{2}\{\gamma^\mu,\gamma^\nu\}P_\mu P_\nu - M^2\Big)\Psi. \tag{5.13}$$

The last step holds because $P_\mu P_\nu$ is symmetric in its two indices, so only the symmetric part of $\gamma^\nu\gamma^\mu$ survives. For this to equal $(P^2 - M^2)\Psi$, we must have

$$\boxed{\{\gamma^\mu,\gamma^\nu\} \equiv \gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2\eta^{\mu\nu}\mathbf{1}.} \tag{5.14}$$

This is called the **Clifford algebra**. Spelled out: $(\gamma^0)^2 = \mathbf{1}$, $(\gamma^i)^2 = -\mathbf{1}$, and different γ's anticommute pairwise. Its origin is precisely Dirac's wish: the square of the first-order equation must be the Klein–Gordon equation.

What is the smallest size such matrices can have? Numbers won't do: numbers commute with each other. 2×2 won't do either: there are at most three pairwise anticommuting 2×2 matrices (the three Pauli matrices), and we need four. The dimension must also be even: from $\gamma^0\gamma^i = -\gamma^i\gamma^0$ and $(\gamma^0)^2 = \mathbf{1}$ we get $\gamma^i = -\gamma^0\gamma^i\gamma^0$, and taking the trace gives $\operatorname{tr}\gamma^i = 0$; meanwhile $(\gamma^i)^2 = -\mathbf{1}$ says its eigenvalues can only be $\pm\mathrm{i}$, and a vanishing trace requires equal numbers of each. So the smallest size is 4×4, and Ψ has four complex components. This Ψ is called a **Dirac spinor**.

5.2.1 The Lagrangian

The Lagrangian that gives the Dirac equation is

$$\boxed{\mathcal{L} = \bar{\Psi}\left(\mathrm{i}\gamma^\mu\partial_\mu - M\right)\Psi, \qquad \bar{\Psi} \equiv \Psi^\dagger\gamma^0.} \tag{5.15}$$

Following the complex scalar field, treat Ψ and $\bar{\Psi}$ as independent variables. $\mathcal{L}$ contains no $\partial\bar{\Psi}$, so the Euler–Lagrange equation for $\bar{\Psi}$ is simply $\partial\mathcal{L}/\partial\bar{\Psi} = 0$:

$$(\mathrm{i}\gamma^\mu\partial_\mu - M)\Psi = 0 \qquad \Longleftrightarrow \qquad (\not{P} - M)\Psi = 0, \tag{5.16}$$

where $\not{P} \equiv \gamma^\mu P_\mu$ (read "P slash"). Substituting a plane wave $\Psi = u\,e^{-\mathrm{i}p\cdot x}$ gives $(\not{p} - M)u = 0$, and multiplying on the left by $(\not{p} + M)$ immediately gives $(p^2 - M^2)u = 0$. The first Casimir equals M^2, just as we wanted.

You might wonder why we use $\bar{\Psi} = \Psi^\dagger\gamma^0$ rather than the more natural $\Psi^\dagger$. The reason is in the next section: the transformation matrix of Ψ is not unitary, so $\Psi^\dagger\Psi$ is not a Lorentz scalar, and inserting a γ^0 fixes exactly that defect.

As usual, let's count dimensions. In D-dimensional spacetime $[\mathcal{L}] = D$, and the kinetic term $\bar{\Psi}\,\partial\Psi$ contains two Ψ's and *one* derivative, so

$$2[\Psi] + 1 = D \qquad \Longrightarrow \qquad [\Psi] = \frac{D-1}{2} \xrightarrow{D=4} \frac{3}{2}. \tag{5.17}$$

The dimension of a spinor field exceeds that of a scalar field by $\frac{1}{2}$, simply because its kinetic term has one fewer derivative. While we're at it, check the mass term: $[M\bar{\Psi}\Psi] = 1 + \frac{3}{2} + \frac{3}{2} = 4$, consistent.[1]

5.3 Lorentz Transformations of the Dirac Field

Does the Dirac equation look the same in every inertial frame? To answer that, we first need to know how Ψ transforms. The answer is the following set of generators:

$$\boxed{\Sigma^{\mu\nu} = \frac{\mathrm{i}}{4}[\gamma^\mu, \gamma^\nu].} \tag{5.18}$$

What qualifies them as Lorentz generators? The key step uses nothing but the Clifford algebra. First compute

$$[\gamma^\mu\gamma^\nu, \gamma^\rho] = \gamma^\mu\{\gamma^\nu, \gamma^\rho\} - \{\gamma^\mu, \gamma^\rho\}\gamma^\nu = 2\eta^{\nu\rho}\gamma^\mu - 2\eta^{\mu\rho}\gamma^\nu$$

[1] In other books you will also see $\mathcal{L} = -\mathrm{i}\bar{\Psi}\gamma^\mu\partial_\mu\Psi - M\bar{\Psi}\Psi$; that is because they take $P_\mu = -\mathrm{i}\partial_\mu$. Written in terms of momentum, both forms are $\bar{\Psi}(\not{P} - M)\Psi$. As for the sign in front of the mass term, it can be flipped at will by the field redefinition $\Psi \to \gamma^5\Psi$ (see Eq. (5.30) for the definition of γ^5), which leaves the kinetic term untouched, so it has no physical meaning.

(expand the middle expression and you'll confirm the first equality). Next notice that $\Sigma^{\mu\nu} = \frac{\mathrm{i}}{2}(\gamma^\mu\gamma^\nu - \eta^{\mu\nu}\mathbf{1})$, and the constant term doesn't affect commutators, so

$$\boxed{[\Sigma^{\mu\nu}, \gamma^\rho] = \mathrm{i}\big(\eta^{\nu\rho}\gamma^\mu - \eta^{\mu\rho}\gamma^\nu\big).} \tag{5.19}$$

Compare this with Eq. (1.29): the behavior of γ^ρ under Σ is identical to the behavior of the four-momentum P^ρ under L; in other words, γ^ρ transforms like a four-vector. Using this result once more, you can show that $\Sigma_{\mu\nu}$ (with indices lowered) satisfies exactly the same commutation relations as Eq. (1.33). So Σ is a 4×4 representation of the Lorentz algebra, with finite transformations

$$S(\Lambda) = \exp\Big(-\frac{\mathrm{i}}{2}\omega^{\mu\nu}\Sigma_{\mu\nu}\Big), \qquad \boxed{S^{-1}\gamma^\mu S = \Lambda^\mu{}_\nu\gamma^\nu,} \tag{5.20}$$

the latter being the finite version of (5.19).

Now we can prove that the Dirac equation is covariant. Let $\Psi'(x') = S\Psi(x)$ with $x' = \Lambda x$; derivatives transform with the inverse matrix, $\partial'_\mu = (\Lambda^{-1})^\nu{}_\mu\partial_\nu$. Then

$$\begin{aligned}(\mathrm{i}\gamma^\mu\partial'_\mu - M)\Psi' &= S\big(\mathrm{i}\,S^{-1}\gamma^\mu S\,(\Lambda^{-1})^\nu{}_\mu\partial_\nu - M\big)\Psi \\ &= S\big(\mathrm{i}\,\Lambda^\mu{}_\rho(\Lambda^{-1})^\nu{}_\mu\gamma^\rho\partial_\nu - M\big)\Psi = S(\mathrm{i}\gamma^\nu\partial_\nu - M)\Psi = 0.\end{aligned} \tag{5.21}$$

If it holds in one inertial frame, it holds in all of them.

Why is $\bar\Psi\Psi$ a scalar? From the Clifford algebra one can check that $\gamma^0\gamma^{\mu\dagger}\gamma^0 = \gamma^\mu$, hence $\Sigma^{\mu\nu\dagger} = \gamma^0\Sigma^{\mu\nu}\gamma^0$, and so $S^\dagger = \gamma^0 S^{-1}\gamma^0$. Therefore

$$\bar\Psi' = \Psi^\dagger S^\dagger\gamma^0 = \Psi^\dagger\gamma^0 S^{-1} = \bar\Psi S^{-1}, \qquad \bar\Psi'\Psi' = \bar\Psi\Psi. \tag{5.22}$$

S is not unitary (its boost part isn't), so $S^\dagger \neq S^{-1}$, but γ^0 turns $S^\dagger$ into exactly S^{-1}. That is where the γ^0 in the definition of $\bar\Psi$ comes from.

Finally, a small technical trick. If all you want to know is how the *components* of Ψ change, look at the origin: $\Psi(0) \to S\Psi(0)$. The origin doesn't move under Lorentz transformations, and the orbital generators $\mathrm{i}(x_\mu\partial_\nu - x_\nu\partial_\mu)$ vanish at $x = 0$, so all that remains is Σ, the pure spin part.

Problem 5.3 In the Weyl representation (5.23) of the next section, use block matrix multiplication to verify directly that $(\gamma^1)^2 = -\mathbf{1}$ and $\{\gamma^0, \gamma^1\} = 0$.

5.4 The Weyl Representation: Splitting a Dirac Spinor in Half

The γ matrices satisfying the Clifford algebra are not unique; any set can be turned into any other by a similarity transformation. The most convenient one for us is the **Weyl representation** (also called the chiral representation):

$$\boxed{\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu \\ \bar\sigma^\mu & 0 \end{pmatrix},} \tag{5.23}$$

where each block is a 2×2 matrix. It satisfies the Clifford algebra:

$$\{\gamma^\mu, \gamma^\nu\} = \begin{pmatrix} \sigma^\mu\bar{\sigma}^\nu + \sigma^\nu\bar{\sigma}^\mu & 0 \\ 0 & \bar{\sigma}^\mu\sigma^\nu + \bar{\sigma}^\nu\sigma^\mu \end{pmatrix} = 2\eta^{\mu\nu}\mathbf{1}, \tag{5.24}$$

using precisely the identity (2.14) from Chapter 2.

The beauty of this representation is that the generators become block diagonal. Since $\gamma^\mu\gamma^\nu = \mathrm{diag}(\sigma^\mu\bar{\sigma}^\nu, \bar{\sigma}^\mu\sigma^\nu)$,

$$\boxed{\Sigma^{\mu\nu} = \begin{pmatrix} \sigma^{\mu\nu} & 0 \\ 0 & \bar{\sigma}^{\mu\nu} \end{pmatrix},} \tag{5.25}$$

where

$$(\sigma^{\mu\nu})_\alpha{}^\beta = \frac{\mathrm{i}}{4}\big(\sigma^\mu\bar{\sigma}^\nu - \sigma^\nu\bar{\sigma}^\mu\big)_\alpha{}^\beta, \qquad (\bar{\sigma}^{\mu\nu})^{\dot{\alpha}}{}_{\dot{\beta}} = \frac{\mathrm{i}}{4}\big(\bar{\sigma}^\mu\sigma^\nu - \bar{\sigma}^\nu\sigma^\mu\big)^{\dot{\alpha}}{}_{\dot{\beta}}. \tag{5.26}$$

The index positions are fixed by matrix multiplication: in $\sigma^\mu_{\alpha\dot{\alpha}}\bar{\sigma}^{\nu\,\dot{\alpha}\beta}$, the middle $\dot{\alpha}$ has been contracted away, leaving α down and β up.

Block diagonal means: the top two components mix only with the top two, and the bottom two only with the bottom two. The four-component Dirac representation is reducible; it splits into two two-component representations:

$$\boxed{\Psi = \begin{pmatrix} \psi_\alpha \\ \bar{\chi}^{\dot{\alpha}} \end{pmatrix} \in \left(\tfrac{1}{2}, 0\right) \oplus \left(0, \tfrac{1}{2}\right).} \tag{5.27}$$

This labeling is what we promised in Chapter 2 to verify, and now we'll compute it.

5.4.1 Left-Handed and Right-Handed

Substitute $\sigma^\mu = (\mathbf{1}, \boldsymbol{\sigma})$ and $\bar{\sigma}^\mu = (\mathbf{1}, -\boldsymbol{\sigma})$ into (5.25):

$$\sigma^{ij} = \frac{\mathrm{i}}{4}\big(-\sigma_i\sigma_j + \sigma_j\sigma_i\big) = \frac{\mathrm{i}}{4}\big(-2\mathrm{i}\varepsilon_{ijk}\sigma_k\big) = \frac{1}{2}\varepsilon_{ijk}\sigma_k = \bar{\sigma}^{ij}, \tag{5.28}$$

$$\sigma^{0i} = \frac{\mathrm{i}}{4}\big(-\sigma_i - \sigma_i\big) = -\frac{\mathrm{i}}{2}\sigma_i, \qquad \bar{\sigma}^{0i} = \frac{\mathrm{i}}{4}\big(\sigma_i + \sigma_i\big) = +\frac{\mathrm{i}}{2}\sigma_i. \tag{5.29}$$

Read off the rotation and boost generators according to Eq. (2.1), and then combine them into J^L and J^R:

	$R_i = \frac{1}{2}\varepsilon_{ijk}\Sigma^{jk}$	$K_i = \Sigma^{0i}$	J_i^L	J_i^R
Upper block ψ_α	$\sigma_i/2$	$-\mathrm{i}\sigma_i/2$	$\sigma_i/2$	0
Lower block $\bar{\chi}^{\dot{\alpha}}$	$\sigma_i/2$	$+\mathrm{i}\sigma_i/2$	0	$\sigma_i/2$

This table deserves a second look. The rotation generators are the *same* in both blocks: the two kinds of spinor are indistinguishable under spatial rotations, both being ordinary spin $\frac{1}{2}$. The boost generators *differ by a sign* between the blocks: under boosts they transform oppositely. That is what **chirality** means. In the upper block only J^L is nonzero, so ψ_α belongs to $(\frac{1}{2}, 0)$; the lower block has only J^R, so $\bar{\chi}^{\dot{\alpha}}$ belongs to $(0, \frac{1}{2})$. Compare with the example in Chapter 2: the R and K of the upper block are exactly the $(\frac{1}{2}, 0)$ generators we found there.

The finite transformations of the upper block are just the M of Chapter 2. Rotation by θ about z: $e^{-\mathrm{i}\theta R_3} = e^{-\mathrm{i}\theta\sigma_3/2} = M_R(\theta)$; boost along z: $e^{-\mathrm{i}\chi K_3} = e^{-\chi\sigma_3/2} = M_B(\chi)$. The lower

block corresponds to $(M^\dagger)^{-1}$: for rotations M_R is unitary and the two blocks are the same; for boosts the lower block is $e^{+\chi\sigma_3/2}$, exactly opposite to the upper block.

The two chiralities can be cleanly separated by a single matrix. Define

$$\gamma^5 \equiv \mathrm{i}\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} -\mathbf{1} & 0 \\ 0 & \mathbf{1} \end{pmatrix}. \tag{5.30}$$

It anticommutes with every γ^μ, squares to $\mathbf{1}$, and is diagonal in the Weyl representation. $P_L = \frac{1}{2}(\mathbf{1} - \gamma^5)$ picks out the upper block ψ_α, and $P_R = \frac{1}{2}(\mathbf{1} + \gamma^5)$ picks out the lower block $\bar\chi^{\dot\alpha}$.

5.4.2 The Lagrangian in Two-Component Form

In the Weyl representation $\gamma^0 = \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix}$, which swaps the upper and lower blocks. $\Psi^\dagger = (\bar\psi_{\dot\alpha}, \chi^\alpha)$ (under complex conjugation ψ_α becomes $\bar\psi_{\dot\alpha}$, and $\bar\chi^{\dot\alpha}$ turns back into χ^α), so

$$\bar\Psi = \Psi^\dagger\gamma^0 = \big(\chi^\alpha,\ \bar\psi_{\dot\alpha}\big). \tag{5.31}$$

Substitute all this into (5.15) and multiply block by block:

$$\boxed{\mathcal{L} = \mathrm{i}\,\bar\psi_{\dot\alpha}\bar\sigma^{\mu\dot\alpha\alpha}\partial_\mu\psi_\alpha + \mathrm{i}\,\chi^\alpha\sigma^\mu_{\alpha\dot\alpha}\partial_\mu\bar\chi^{\dot\alpha} - M\big(\chi^\alpha\psi_\alpha + \bar\psi_{\dot\alpha}\bar\chi^{\dot\alpha}\big).} \tag{5.32}$$

The indices in every term are contracted according to the convention (5.7), so each term on its own is Lorentz invariant. Read this formula carefully: in the kinetic terms ψ pairs only with $\bar\psi$, and χ only with $\bar\chi$; the two chiralities go their own ways, and only the mass term ties them together. An important conclusion follows at once: *without mass, the Dirac field falls apart into two completely unrelated two-component fields.* The next section starts from here.

Example 5.1 How many spin states does a Dirac particle at rest have?

Find the positive-energy solutions at rest, and verify that they describe spin $\frac{1}{2}$.

Solution At rest $\Psi = u\,e^{-\mathrm{i}Mt}$, and the Dirac equation (5.16) becomes $(M\gamma^0 - M)u = 0$, i.e., $\gamma^0 u = u$. γ^0 swaps the upper and lower blocks, so the two blocks must be equal:

$$u = \begin{pmatrix} \xi \\ \xi \end{pmatrix}, \qquad \xi = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ or } \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Exactly two independent solutions. The rotation generators are $\sigma_i/2$ on both blocks, so R_3 takes the values $+\frac{1}{2}$ and $-\frac{1}{2}$ on these two solutions, and $\boldsymbol{R}^2 = \frac{3}{4} = \frac{1}{2}(\frac{1}{2}+1)$. By Eq. (3.14), $C_2 = -M^2\cdot\frac{3}{4}$, i.e., $s = \frac{1}{2}$, and the number of states is exactly $2s+1 = 2$. □

Problem 5.4 In the Weyl representation, verify that $(\gamma^5)^2 = \mathbf{1}$ and that γ^5 anticommutes with every γ^μ. (There's no need to actually multiply out the four matrices: use the block structure.)

***Problem 5.5** Find the negative-energy solutions $\Psi = v\,e^{+\mathrm{i}Mt}$ of the Dirac equation at rest. What condition does v satisfy? How many independent solutions are there? (After quantization, they describe antiparticles.)

5.5 The Weyl Equation and Helicity

Set $M = 0$ in (5.32), and ψ and χ go their separate ways completely; we are perfectly free to keep just one of them:

$$\boxed{\mathcal{L} = \mathrm{i}\,\bar{\psi}_{\dot{\alpha}}\bar{\sigma}^{\mu\dot{\alpha}\alpha}\partial_\mu\psi_\alpha.} \tag{5.33}$$

This is a field with only two complex components. Varying with respect to $\bar{\psi}$ gives the **Weyl equation**:

$$\boxed{\bar{\sigma}^{\mu\dot{\alpha}\alpha}\partial_\mu\psi_\alpha = 0.} \tag{5.34}$$

Why is this Lagrangian Lorentz invariant? Regard $\bar{\psi}$ as a row vector; it becomes $\bar{\psi}M^\dagger$. By (5.11),

$$\bar{\psi}\bar{\sigma}^\mu\psi \;\to\; \bar{\psi}\,(M^\dagger\bar{\sigma}^\mu M)\,\psi = \Lambda^\mu{}_\nu\,\bar{\psi}\bar{\sigma}^\nu\psi. \tag{5.35}$$

It is a four-vector, and contracted with ∂_μ, which transforms with the inverse matrix, it gives a scalar.

5.5.1 Weyl Particles Are Massless

Act on (5.34) once more with $\sigma^\nu\partial_\nu$, and use the symmetry of $\partial_\mu\partial_\nu$ and the identity (2.14):

$$\sigma^\nu\partial_\nu\,\bar{\sigma}^\mu\partial_\mu\psi = \tfrac{1}{2}\big(\sigma^\nu\bar{\sigma}^\mu + \sigma^\mu\bar{\sigma}^\nu\big)\partial_\mu\partial_\nu\psi = \Box\psi = 0. \tag{5.36}$$

Every component satisfies the *massless* Klein–Gordon equation. This agrees with the conclusion of the last section: with only one chirality, you simply cannot write down a Dirac mass term.

5.5.2 Helicity

Substitute a plane wave $\psi = u\,e^{-\mathrm{i}p\cdot x}$, $p^\mu = (E, \boldsymbol{p})$. $\mathrm{i}\partial_\mu$ becomes $p_\mu = (E, -\boldsymbol{p})$, and

$$p_\mu\bar{\sigma}^\mu = p_0\bar{\sigma}^0 + p_i\bar{\sigma}^i = E\,\mathbf{1} + (-p^i)(-\sigma_i) = E\,\mathbf{1} + \boldsymbol{p}\cdot\boldsymbol{\sigma} \tag{5.37}$$

(the two minus signs from $\bar{\sigma}^i = -\sigma_i$ and $p_i = -p^i$ cancel). The Weyl equation becomes

$$(E + \boldsymbol{p}\cdot\boldsymbol{\sigma})\,u = 0. \tag{5.38}$$

The condition for a nonzero solution is $\det(E + \boldsymbol{p}\cdot\boldsymbol{\sigma}) = E^2 - \boldsymbol{p}^2 = 0$, which shows once again that Weyl particles are massless. Taking $E = |\boldsymbol{p}| > 0$, the equation reduces to

$$(\hat{\boldsymbol{p}}\cdot\boldsymbol{\sigma})\,u = -u. \tag{5.39}$$

The angular momentum on spinors is $\boldsymbol{R} = \boldsymbol{\sigma}/2$, so the helicity is $h = \hat{\boldsymbol{p}}\cdot\boldsymbol{R} = -\frac{1}{2}$: *the spin always points opposite to the direction of motion.* A concrete example: with $\boldsymbol{p}$ along $+z$, the equation is $\mathrm{diag}(2E, 0)\,u = 0$, with solution $u \propto (0,1)^{\mathsf{T}}$: spin down, particle flying up.

The calculation for a right-handed spinor is exactly the same. χ satisfies $\sigma^\mu\partial_\mu\bar{\chi} = 0$, i.e., $(E - \boldsymbol{p}\cdot\boldsymbol{\sigma})v = 0$, which gives $h = +\frac{1}{2}$.

This is a concrete realization of the massless little group of Chapter 3. A left-handed Weyl field gives a particle of helicity $-\frac{1}{2}$; after quantization, its antiparticle has helicity $+\frac{1}{2}$. For quite a long time people believed the neutrino was exactly this kind of particle: experimentally, only left-handed neutrinos and right-handed antineutrinos had ever been seen.[2]

[2] We now know that neutrinos have a tiny mass, so strictly speaking they are not pure Weyl particles. But at energies far above their mass, the Weyl description is still an excellent approximation.

Problem 5.6 A massless left-handed Weyl particle moves in the $+x$ direction, $p^\mu = (E, E, 0, 0)$. Solve $(E + \boldsymbol{p}\cdot\boldsymbol{\sigma})u = 0$ and verify that the helicity is $-\frac{1}{2}$.

Chapter 6

Grassmann Numbers and Majorana Spinors

In the last chapter we wrote down the Dirac and Weyl Lagrangians, but kept dodging one question: what kind of "numbers" are the components of a spinor field, really? For the electron the answer is a little surprising: they are not ordinary numbers, but numbers that *anticommute* with one another. This chapter first makes that clear, and then looks at a few of its consequences: what can be built out of two spinors, and a special kind of spinor, the Majorana spinor, which is its own antiparticle. The generators of supersymmetry are precisely spinors of this kind, so the material of this chapter will be used constantly later on.

6.1 Grassmann Numbers

The spin–statistics theorem tells us that fields of half-integer spin must be quantized with *anti*commutation relations; otherwise the theory goes wrong, either with energy unbounded below or with a violation of causality. The Pauli exclusion principle (no two identical fermions can occupy the same state) is a direct consequence of this. When writing classical Lagrangians, we treat the components of fermion fields as **Grassmann numbers**:

$$\boxed{\psi_\alpha \chi_\beta = -\chi_\beta \psi_\alpha.} \tag{6.1}$$

In particular, taking $\chi = \psi$ and $\beta = \alpha$ gives $\psi_1\psi_1 = -\psi_1\psi_1$, so $\psi_1\psi_1 = 0$. The square of a Grassmann number is zero. Grassmann numbers and ordinary numbers (including the matrix elements of ε and σ) can be moved past each other freely.

6.1.1 A Conclusion That Looks Strange at First

Since the components anticommute, you would probably guess that $\psi\chi = -\chi\psi$. Wrong. Let's work it out step by step:

$$\begin{aligned}
\psi\chi = \psi^\alpha \chi_\alpha &= \varepsilon^{\alpha\beta}\psi_\beta\chi_\alpha && \text{(raise the index)}\\
&= -\varepsilon^{\beta\alpha}\psi_\beta\chi_\alpha && (\varepsilon \text{ is antisymmetric})\\
&= -\varepsilon^{\alpha\beta}\psi_\alpha\chi_\beta && \text{(rename dummy indices)}\\
&= +\varepsilon^{\alpha\beta}\chi_\beta\psi_\alpha && \text{(Grassmann swap)}\\
&= \chi^\alpha\psi_\alpha = \chi\psi. &&
\end{aligned} \tag{6.2}$$

Two minus signs, one from the antisymmetry of ε and one from Grassmann anticommutation, multiply to a plus:

$$\boxed{\psi\chi = \chi\psi, \qquad \bar\psi\bar\chi = \bar\chi\bar\psi.} \tag{6.3}$$

The dotted case is exactly the same.

This conclusion has an immediate and nontrivial consequence. Taking $\chi = \psi$, by (5.8),

$$\psi\psi = \psi_2\psi_1 - \psi_1\psi_2 = 2\psi_2\psi_1. \tag{6.4}$$

If the ψ_α were ordinary numbers, $\psi\psi$ would vanish identically because of the antisymmetry of ε; it is only because the components are Grassmann numbers that it is nonzero. Remember this formula: the Majorana mass term of Section 6.3 depends on it entirely.

6.1.2 Two Useful Identities

When taking the Hermitian conjugate of Grassmann numbers, the order of a product is reversed, $(\theta_1\theta_2)^\dagger = \theta_2^\dagger\theta_1^\dagger$, just as for the Hermitian conjugate of matrices. So

$$(\psi\chi)^\dagger = (\psi^\alpha\chi_\alpha)^\dagger = \bar\chi_{\dot\alpha}\bar\psi^{\dot\alpha} = \bar\chi\bar\psi = \bar\psi\bar\chi. \tag{6.5}$$

Thus the two terms of the Dirac mass term $M(\chi\psi + \bar\psi\bar\chi)$ are Hermitian conjugates of each other, and the Lagrangian is real.

The other identity concerns swapping the positions of the two spinors in a vector bilinear:

$$\boxed{\chi\sigma^\mu\bar\psi = -\bar\psi\bar\sigma^\mu\chi.} \tag{6.6}$$

Here is the derivation. First swap using the Grassmann property: $\chi^\alpha\sigma^\mu_{\alpha\dot\alpha}\bar\psi^{\dot\alpha} = -\bar\psi^{\dot\alpha}\sigma^\mu_{\alpha\dot\alpha}\chi^\alpha$. Then use (5.10) backwards, $\sigma^\mu_{\alpha\dot\alpha} = \varepsilon_{\alpha\beta}\varepsilon_{\dot\alpha\dot\beta}\bar\sigma^{\mu\dot\beta\beta}$, and note that $\bar\psi^{\dot\alpha}\varepsilon_{\dot\alpha\dot\beta} = -\bar\psi_{\dot\beta}$ and $\varepsilon_{\alpha\beta}\chi^\alpha = -\chi_\beta$. These two minus signs cancel each other, leaving only the one from the Grassmann swap.

It has a pretty application. In (6.6), let $\bar\psi \to \partial_\mu\bar\chi$:

$$\chi\sigma^\mu\partial_\mu\bar\chi = -(\partial_\mu\bar\chi)\bar\sigma^\mu\chi = \bar\chi\bar\sigma^\mu\partial_\mu\chi - \partial_\mu(\bar\chi\bar\sigma^\mu\chi).$$

The last term is a total derivative and contributes nothing to the action. So the Dirac Lagrangian (5.32) can be written

$$\mathcal{L} \simeq \mathrm{i}\,\bar\psi\bar\sigma^\mu\partial_\mu\psi + \mathrm{i}\,\bar\chi\bar\sigma^\mu\partial_\mu\chi - M(\chi\psi + \bar\psi\bar\chi), \tag{6.7}$$

where $\simeq$ means "equal up to a total derivative". Now ψ and χ stand on a completely equal footing: *a Dirac fermion is just two left-handed Weyl spinors, plus a mass term that links them.* The supersymmetry literature almost always uses this "everything left-handed" form.

6.2 What Can Be Built from Two Spinors

As we said in Chapter 2, Lorentz representations are labeled by two independent copies of angular momentum (j_L, j_R). So when two representations are multiplied, the left and right copies *each* add according to the ordinary rules of angular momentum addition:

$$(j_L, j_R) \otimes (j'_L, j'_R) = \bigoplus (j''_L, j''_R), \tag{6.8}$$

where j''_L runs over $|j_L - j'_L|, \ldots, j_L + j'_L$ and j''_R over $|j_R - j'_R|, \ldots, j_R + j'_R$. The two most common examples:

$$\left(\tfrac12, 0\right) \otimes \left(0, \tfrac12\right) = \left(\tfrac12, \tfrac12\right), \tag{6.9}$$

$$\left(\tfrac12, 0\right) \otimes \left(\tfrac12, 0\right) = (0,0) \oplus (1,0). \tag{6.10}$$

The first says that a left-handed spinor and a right-handed spinor combine into a four-vector; the explicit combination is $\chi\sigma^\mu\bar\psi$. Under spatial rotations it contains spin 1 (the three spatial components) and spin 0 (the time component). The second is just the familiar $\frac12 \otimes \frac12 = 0 \oplus 1$ of quantum mechanics (singlet plus triplet), only happening in the J^L copy; and the dimensions match, $2 \times 2 = 4 = 1 + 3$.

6.2.1 The Fierz Decomposition

The second example can be made very concrete. $\psi_\alpha\chi_\beta$ has four components, and like any 2×2 matrix it can be split into an antisymmetric part and a symmetric part. The antisymmetric part has only one independent component, so it must be proportional to $\varepsilon_{\alpha\beta}$; it is the $(0,0)$, and its coefficient can be fixed by taking $\alpha=1,\beta=2$. The antisymmetric part is $\frac{1}{2}(\psi_1\chi_2-\psi_2\chi_1)$, and by (5.8) $\psi\chi=\psi_2\chi_1-\psi_1\chi_2$, while $\varepsilon_{12}=-1$, so it equals $\frac{1}{2}\varepsilon_{\alpha\beta}(\psi\chi)$. The symmetric part has three components; it is the $(1,0)$, and it can be expanded in terms of $\sigma^{\mu\nu}$, because the matrix $(\sigma^{\mu\nu}\varepsilon)_{\alpha\beta}$ happens to be symmetric in $\alpha\beta$. Working out the coefficients (this step is rather tedious, so we just quote the result), we get

$$\boxed{\psi_\alpha\chi_\beta=\frac{1}{2}\varepsilon_{\alpha\beta}(\psi\chi)+\frac{1}{2}(\psi\sigma_{\mu\nu}\chi)\,(\sigma^{\mu\nu})_\alpha{}^\gamma\,\varepsilon_{\beta\gamma},} \tag{6.11}$$

where $\psi\sigma_{\mu\nu}\chi\equiv\psi^\alpha(\sigma_{\mu\nu})_\alpha{}^\beta\chi_\beta$. This is called the **Fierz identity**. Throughout the formula ψ always stays to the left of χ, so it holds for Grassmann components and ordinary components alike.[1]

6.2.2 Self-Duality

A general antisymmetric tensor $T_{\mu\nu}$ has six components, yet $\psi\sigma^{\mu\nu}\chi$ has only three independent components. Why is that? The reason is that $\sigma^{\mu\nu}$ obeys a constraint, called the **self-duality** condition (remember our convention $\varepsilon^{0123}=-1$):

$$\boxed{\frac{1}{2}\varepsilon^{\mu\nu\rho\sigma}\sigma_{\rho\sigma}=-\mathrm{i}\,\sigma^{\mu\nu},\qquad\frac{1}{2}\varepsilon^{\mu\nu\rho\sigma}\bar\sigma_{\rho\sigma}=+\mathrm{i}\,\bar\sigma^{\mu\nu}.} \tag{6.12}$$

Let's check one component: take $\mu\nu=01$. The left side is $\varepsilon^{0123}\sigma_{23}=-\sigma^{23}=-\frac{1}{2}\sigma_1$; the right side is $-\mathrm{i}\sigma^{01}=-\mathrm{i}\cdot(-\frac{\mathrm{i}}{2}\sigma_1)=-\frac{1}{2}\sigma_1$. The two sides agree.

Doing the "dual" operation $T_{\mu\nu}\mapsto\frac{1}{2}\varepsilon_{\mu\nu}{}^{\rho\sigma}T_{\rho\sigma}$ twice amounts to multiplying by -1, so its eigenvalues are $\pm\mathrm{i}$. The six-dimensional space of antisymmetric tensors therefore splits into two three-dimensional subspaces: the one with eigenvalue $-\mathrm{i}$ is $(1,0)$, and the one with $+\mathrm{i}$ is $(0,1)$. $\psi\sigma^{\mu\nu}\chi$ lands only in the first. In Chapter 7 we will see that the electromagnetic field strength $F_{\mu\nu}$ splits in two in exactly this way.

***Problem 6.1** Let θ_α be a Grassmann two-component spinor. Use the Fierz identity to show that $\theta_\alpha\theta_\beta=\frac{1}{2}\varepsilon_{\alpha\beta}(\theta\theta)$, and check it directly for $\alpha=1,\beta=2$. (This little identity will be used hundreds of times in superspace.)

Problem 6.2 Verify the self-duality condition (6.12) for $\mu\nu=12$.

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[1] The index order on that last ε is very easy to get backwards, and if you do, the second term is off by a minus sign. Moreover, different books use different conventions for ε and $\sigma^{\mu\nu}$; whenever you copy an identity of this kind from somewhere else, be sure to check the conventions first.

6.3 Charge Conjugation and Majorana Spinors

6.3.1 Charge Conjugation

Suppose the Dirac field is coupled to the electromagnetic field, with equation of motion

$$\left[\mathrm{i}\gamma^\mu(\partial_\mu + \mathrm{i}eA_\mu) - M\right]\Psi = 0. \tag{6.13}$$

We want to build from Ψ a new field Ψ^c that satisfies the same equation, but with the charge e replaced by $-e$. If we can find one, Ψ^c describes the antiparticle. The method takes three steps. First take the Hermitian conjugate (A_μ is real):

$$-\mathrm{i}(\partial_\mu\Psi^\dagger)\gamma^{\mu\dagger} - eA_\mu\Psi^\dagger\gamma^{\mu\dagger} - M\Psi^\dagger = 0.$$

Next multiply on the right by γ^0, and use $\gamma^{\mu\dagger}\gamma^0 = \gamma^0\gamma^\mu$ to write it as an equation for $\bar\Psi$:

$$-\mathrm{i}(\partial_\mu\bar\Psi)\gamma^\mu - eA_\mu\bar\Psi\gamma^\mu - M\bar\Psi = 0.$$

Finally transpose, and multiply on the left by a matrix C. If C satisfies

$$C\gamma^{\mu\mathsf{T}}C^{-1} = -\gamma^\mu, \tag{6.14}$$

then $\Psi^c \equiv C\bar\Psi^{\mathsf{T}}$ satisfies

$$\left[\mathrm{i}\gamma^\mu(\partial_\mu - \mathrm{i}eA_\mu) - M\right]\Psi^c = 0. \tag{6.15}$$

The charge has indeed flipped sign. C is called the **charge conjugation matrix**. In the Weyl representation we can take

$$\boxed{C = \mathrm{i}\gamma^0\gamma^2 = \begin{pmatrix} -\mathrm{i}\sigma_2 & 0 \\ 0 & \mathrm{i}\sigma_2 \end{pmatrix} = \begin{pmatrix} \varepsilon_{\alpha\beta} & 0 \\ 0 & \varepsilon^{\dot\alpha\dot\beta} \end{pmatrix}.} \tag{6.16}$$

For the last step just compare with (5.2): $-\mathrm{i}\sigma_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \varepsilon_{\alpha\beta}$ and $\mathrm{i}\sigma_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \varepsilon^{\dot\alpha\dot\beta}$. To put it bluntly, the upper block of C is "lower an undotted index", and the lower block is "raise a dotted index". (It satisfies (6.14); you can check each case.)

Computing Ψ^c with two-component spinors makes everything transparent. In the last chapter we found $\bar\Psi = (\chi^\alpha, \bar\psi_{\dot\alpha})$; transpose it into a column vector and multiply on the left by C:

$$\Psi = \begin{pmatrix} \psi_\alpha \\ \bar\chi^{\dot\alpha} \end{pmatrix} \quad\longrightarrow\quad \Psi^c = C\bar\Psi^{\mathsf{T}} = \begin{pmatrix} \varepsilon_{\alpha\beta}\chi^\beta \\ \varepsilon^{\dot\alpha\dot\beta}\bar\psi_{\dot\beta} \end{pmatrix} = \boxed{\begin{pmatrix} \chi_\alpha \\ \bar\psi^{\dot\alpha} \end{pmatrix}}. \tag{6.17}$$

Charge conjugation does nothing more than swap ψ and χ! Do it again and they swap back, so $(\Psi^c)^c = \Psi$.

6.3.2 The Majorana Condition

Now a natural question: is there a kind of spinor whose charge conjugate is itself?

$$\boxed{\Psi^c = \Psi \quad\Longleftrightarrow\quad \chi = \psi \quad\Longleftrightarrow\quad \Psi_M = \begin{pmatrix} \psi_\alpha \\ \bar\psi^{\dot\alpha} \end{pmatrix}.} \tag{6.18}$$

This is called the **Majorana condition**, and a spinor that satisfies it is called a Majorana spinor. It is entirely analogous to "a complex scalar φ satisfying $\varphi^* = \varphi$ becomes a real scalar", so it is also called a reality condition. A Majorana particle is its own antiparticle, so it cannot carry any $U(1)$ charge, electric charge for instance.

Count the degrees of freedom: a Dirac spinor has 4 complex components, i.e., 8 real components; a Majorana spinor is left with only the 2 complex components of ψ_α, i.e., 4 real components, the same as a Weyl spinor. A four-component Majorana spinor and a two-component Weyl spinor carry exactly the same information; they are just written differently.

6.3.3 The Majorana Mass

In that case, what difference is there between a Majorana spinor and a Weyl spinor? The difference lies in the mass. Substitute $\chi = \psi$ into the Dirac Lagrangian, multiply by $\frac{1}{2}$ (for the same reason as the $\frac{1}{2}$ in the real scalar Lagrangian: Ψ_M and $\bar{\Psi}_M$ are no longer independent, and without it we would be double counting), and tidy up with the total-derivative identity of the previous section. We get

$$\boxed{\mathcal{L} = \frac{1}{2}\bar{\Psi}_M(\mathrm{i}\gamma^\mu\partial_\mu - M)\Psi_M \simeq \mathrm{i}\,\bar{\psi}\bar{\sigma}^\mu\partial_\mu\psi - \frac{M}{2}\big(\psi\psi + \bar{\psi}\bar{\psi}\big).} \tag{6.19}$$

In the last chapter we said that with only one chirality you cannot write the Dirac mass term $\chi\psi$, and that's true; but we can write $\psi\psi$! It is nonzero entirely thanks to the Grassmann property $\psi\psi = 2\psi_2\psi_1$. This is called a **Majorana mass**. The price: under the phase transformation $\psi \to e^{\mathrm{i}\alpha}\psi$, $\psi\psi \to e^{2\mathrm{i}\alpha}\psi\psi$, so particles that carry a $U(1)$ charge cannot have a Majorana mass. This is the same statement as "a Majorana particle is its own antiparticle".

Why does supersymmetry care so much about Majorana spinors? Because the supersymmetry generators Q_α and $\bar{Q}_{\dot{\alpha}} = (Q_\alpha)^\dagger$ are precisely a pair of two-component spinors that are complex conjugates of each other, and together they form a Majorana spinor. The superpartners of the gauge bosons (the gauginos) are Majorana fermions too. So the two-component language of this chapter will be our everyday language later on.

Problem 6.3 Use Eq. (6.17) to show that $(\Psi^c)^c = \Psi$. Why can't the Majorana condition be imposed on a charged Dirac field?

****Problem 6.4** Starting from $\frac{1}{2}\bar{\Psi}_M(\mathrm{i}\gamma^\mu\partial_\mu - M)\Psi_M$, and using $\bar{\Psi}_M = (\psi^\alpha, \bar{\psi}_{\dot{\alpha}})$ together with Eq. (6.6), derive the right-hand side of Eq. (6.19).

Chapter 7

Vector Fields

The last kind of field we will study is the vector field, whose ancestor is the electromagnetic four-potential. Vector fields have a new feature that spinor fields lack: some of their four components are "fake", a mere redundancy in the description that corresponds to no physical state at all. Sorting out which components are real and which are fake is the main task of this chapter. The answer will agree perfectly with the prediction of Chapter 3 for massless particles: the photon has just two states, helicity ± 1.

7.1 The Maxwell Field

A vector field assigns a four-vector to every spacetime point, and under a transformation its components mix just like coordinate differences do:

$$\boxed{A'^{\mu}(x') = \Lambda^{\mu}{}_{\nu} A^{\nu}(x).} \tag{7.1}$$

It belongs to the $(\frac{1}{2}, \frac{1}{2})$ representation. Under spatial rotations, A^0 is spin 0 and $\boldsymbol{A}$ is spin 1. The most familiar example is the electromagnetic four-potential $A^{\mu} = (\phi, \boldsymbol{A})$, from which we get the electric and magnetic fields:

$$\boldsymbol{E} = -\nabla\phi - \partial_t \boldsymbol{A}, \qquad \boldsymbol{B} = \nabla \times \boldsymbol{A}. \tag{7.2}$$

It is more convenient to package them into an antisymmetric tensor. Define the **field strength**

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \tag{7.3}$$

It has six independent components, which are exactly $\boldsymbol{E}$ and $\boldsymbol{B}$:

$$F_{0i} = E^i, \qquad F_{ij} = -\varepsilon_{ijk} B^k. \tag{7.4}$$

For example, $F_{01} = \partial_0 A_1 - \partial_1 A_0 = -\partial_t A^1 - \partial_x \phi = E^1$ (don't forget that $A_1 = -A^1$).

7.1.1 The Lagrangian and the Equations of Motion

The Lagrangian for the electromagnetic field coupled to an external current J^{μ} is

$$\boxed{\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - e J^{\mu} A_{\mu}.} \tag{7.5}$$

The first term looks a bit mysterious, but substitute (7.4) and you'll find it is nothing but $\frac{1}{2}(\boldsymbol{E}^2 - \boldsymbol{B}^2)$ (Problem 7.2): the "kinetic energy" of the electric field minus the "potential energy" of the magnetic field.

Now treat the four components A_ν as four independent fields, and use the Euler–Lagrange equation (4.8). We need two partial derivatives. The first: $\partial F_{\rho\sigma}/\partial(\partial_\mu A_\nu) = \delta^\mu_\rho\delta^\nu_\sigma - \delta^\mu_\sigma\delta^\nu_\rho$, so

$$\frac{\partial(F_{\rho\sigma}F^{\rho\sigma})}{\partial(\partial_\mu A_\nu)} = 2F^{\rho\sigma}\left(\delta^\mu_\rho\delta^\nu_\sigma - \delta^\mu_\sigma\delta^\nu_\rho\right) = 4F^{\mu\nu}, \qquad \frac{\partial\mathcal{L}}{\partial(\partial_\mu A_\nu)} = -F^{\mu\nu}.$$

The second is much simpler: $\partial\mathcal{L}/\partial A_\nu = -eJ^\nu$. Plugging in gives $-\partial_\mu F^{\mu\nu} + eJ^\nu = 0$, that is,

$$\boxed{\partial^\mu F_{\mu\nu} = e\,J_\nu.} \tag{7.6}$$

Look at it component by component:

$$\nu = 0: \quad \partial_i F^{i0} = \nabla\cdot\boldsymbol{E} = e\rho, \tag{7.7}$$

$$\nu = j: \quad \partial_0 F^{0j} + \partial_i F^{ij} = \left(\nabla\times\boldsymbol{B} - \partial_t\boldsymbol{E}\right)^j = eJ^j. \tag{7.8}$$

These are two of Maxwell's equations. The other two ($\nabla\cdot\boldsymbol{B} = 0$ and Faraday's law) don't need a Lagrangian; they are guaranteed automatically by the definition $F = \partial A - \partial A$.

7.2 The Photon Has Only Two Polarizations

A_μ has four components, but the photon has only two polarization states. Where did the other two go?

The first one is eaten by **gauge symmetry**. For an arbitrary function $\alpha(x)$, the transformation

$$A_\mu \to A_\mu + \partial_\mu\alpha \tag{7.9}$$

doesn't change $F_{\mu\nu}$, because $\partial_\mu\partial_\nu\alpha - \partial_\nu\partial_\mu\alpha = 0$. The Lagrangian and all observables depend on A only through F, so they are all unchanged. This means one component of A_μ is pure redundancy in the description: two A_μ's that differ by a gauge transformation describe the same physical state. We can always remove it by "choosing a gauge".

The second one is eaten by a **constraint**. Notice that $F_{00} = 0$, so the Lagrangian contains no $\partial_t A_0$ at all. A_0 has no "velocity" of its own; it is not an independently evolving degree of freedom. The corresponding equation of motion, $\nu = 0$, i.e., Gauss's law $\nabla\cdot\boldsymbol{E} = e\rho$, contains no second time derivatives; it is not an evolution equation but a constraint: at each instant it determines A_0 in terms of the other fields. Remove these two from the four components, and two are left.

Plane waves make this clearest. Choose the Lorenz gauge $\partial_\mu A^\mu = 0$, in which the free equation of motion reduces to $\Box A^\mu = 0$. Substituting $A^\mu = \epsilon^\mu e^{-\mathrm{i}k\cdot x}$ gives $k^2 = 0$ and $k\cdot\epsilon = 0$. Take $k^\mu = (\kappa, 0, 0, \kappa)$; the condition $k\cdot\epsilon = 0$ gives $\epsilon^0 = \epsilon^3$, leaving three. But the Lorenz gauge does not use up all the gauge freedom: gauge transformations with $\Box\alpha = 0$ are still allowed, and they turn ϵ^μ into $\epsilon^\mu + c\,k^\mu$, which can be used to eliminate ϵ^0 (along with ϵ^3). In the end all that's left is

$$\epsilon^\mu = (0,1,0,0) \quad \text{and} \quad (0,0,1,0),$$

two transverse polarizations. Combine them into

$$\epsilon^\mu_\pm = \frac{1}{\sqrt{2}}(0, 1, \pm\mathrm{i}, 0); \tag{7.10}$$

the generator of rotations about the z axis, R_3, acts on the spatial components as the matrix $\begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}$, and acting on $(1, \pm\mathrm{i})$ it gives $\pm(1, \pm\mathrm{i})$. So $\epsilon_\pm$ have helicity $h = \pm 1$, and there is no $h = 0$. This is exactly the prediction of Chapter 3 for massless particles.

If we add a mass term $\frac{1}{2}m^2 A_\mu A^\mu$ to the vector field (this is called the Proca field), things are different. The equation of motion becomes $\partial_\mu F^{\mu\nu} + m^2 A^\nu = 0$; take ∂_ν of it, and the first term vanishes because F is antisymmetric, giving the constraint $m^2 \partial_\nu A^\nu = 0$. The mass term breaks gauge symmetry, so this time only one component is removed, leaving three: exactly the $2s + 1 = 3$ states of a massive spin-1 particle.

Problem 7.1 For $k^\mu = (\kappa, 0, 0, \kappa)$, verify that $\epsilon_\pm^\mu = (0, 1, \pm\mathrm{i}, 0)/\sqrt{2}$ satisfy $k \cdot \epsilon_\pm = 0$. Then explain why $\epsilon^\mu \propto k^\mu$ is not a physical polarization.

Problem 7.2 Using $F_{0i} = E^i$ and $F_{ij} = -\varepsilon_{ijk} B^k$, verify that $-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} = \frac{1}{2}(\boldsymbol{E}^2 - \boldsymbol{B}^2)$.

***Problem 7.3** For a plane wave of the Proca field, $A^\mu = \epsilon^\mu e^{-\mathrm{i}k\cdot x}$, with $k^\mu = (E, 0, 0, k)$ and $E^2 = k^2 + m^2$, find all the independent polarization vectors. Why can't one more be removed this time, as for the photon?

7.3 The Chiral Decomposition of the Field Strength

In Chapter 6 we saw that the six-dimensional space of antisymmetric tensors splits into two halves, the self-dual $(1, 0)$ and the anti-self-dual $(0, 1)$. The electromagnetic field strength is just such an antisymmetric tensor, so it too should split in two. Explicitly,

$$\boxed{F_{\mu\nu} = \frac{1}{2}(\sigma_{\mu\nu})_\alpha{}^\beta F_\beta{}^\alpha + \frac{1}{2}(\bar{\sigma}_{\mu\nu})^{\dot{\alpha}}{}_{\dot{\beta}}\, \bar{F}^{\dot{\beta}}{}_{\dot{\alpha}},} \tag{7.11}$$

where the two "field strengths in spinor form" are defined by

$$F_\alpha{}^\beta \equiv (\sigma^{\rho\sigma})_\alpha{}^\beta F_{\rho\sigma}, \qquad \bar{F}^{\dot{\alpha}}{}_{\dot{\beta}} \equiv (\bar{\sigma}^{\rho\sigma})^{\dot{\alpha}}{}_{\dot{\beta}} F_{\rho\sigma}. \tag{7.12}$$

Verifying (7.11) takes only two trace identities:

$$\operatorname{tr}\left(\sigma^{\mu\nu}\sigma^{\rho\sigma}\right) = \tfrac{1}{2}\left(\eta^{\mu\rho}\eta^{\nu\sigma} - \eta^{\mu\sigma}\eta^{\nu\rho}\right) + \tfrac{\mathrm{i}}{2}\varepsilon^{\mu\nu\rho\sigma}, \tag{7.13}$$

$$\operatorname{tr}\left(\bar{\sigma}^{\mu\nu}\bar{\sigma}^{\rho\sigma}\right) = \tfrac{1}{2}\left(\eta^{\mu\rho}\eta^{\nu\sigma} - \eta^{\mu\sigma}\eta^{\nu\rho}\right) - \tfrac{\mathrm{i}}{2}\varepsilon^{\mu\nu\rho\sigma}. \tag{7.14}$$

Substitute the definitions into the right side of (7.11): the ε terms in the two traces come with opposite signs and cancel when added; the remaining $\eta\eta$ terms reproduce the antisymmetric $F_{\rho\sigma}$ exactly as it was: $\frac{1}{2} \cdot 2 \cdot \frac{1}{2}(F_{\mu\nu} - F_{\nu\mu}) = F_{\mu\nu}$. The fact that the ε terms in the two traces have opposite signs is really just the self-duality condition (6.12) said another way.

Substituting (7.4), we get a very pretty result:

$$\boxed{F_\alpha{}^\beta = -\mathrm{i}\big[(\boldsymbol{E} - \mathrm{i}\boldsymbol{B}) \cdot \boldsymbol{\sigma}\big]_\alpha{}^\beta, \qquad \bar{F}^{\dot{\alpha}}{}_{\dot{\beta}} = +\mathrm{i}\big[(\boldsymbol{E} + \mathrm{i}\boldsymbol{B}) \cdot \boldsymbol{\sigma}\big]^{\dot{\alpha}}{}_{\dot{\beta}}.} \tag{7.15}$$

$F_\alpha{}^\beta$ depends only on the three complex numbers $\boldsymbol{E} - \mathrm{i}\boldsymbol{B}$, and belongs to $(1, 0)$; $\bar{F}$ depends only on their complex conjugate $\boldsymbol{E} + \mathrm{i}\boldsymbol{B}$, and belongs to $(0, 1)$. Together that is exactly six real

numbers, namely $\boldsymbol{E}$ and $\boldsymbol{B}$, not one more and not one less. These two complex combinations carry the photon's two helicities respectively. In supersymmetric theories, the gauge field appears inside superfields precisely in this two-component form $F_{\alpha\beta}$.

7.4 Epilogue: The Ledger of Degrees of Freedom

Along the way we have built a whole set of free fields. Let's put them in one table:

Field	Lorentz representation	Off shell	On shell	Physical states
Real scalar ϕ	$(0,0)$	1	1	spin 0
Complex scalar φ	$(0,0)$	2	2	spin-0 particle and antiparticle
Weyl spinor ψ_α	$(\frac{1}{2},0)$	4	2	$h=-\frac{1}{2}$ and antiparticle $h=+\frac{1}{2}$
Majorana spinor	$(\frac{1}{2},0)\oplus(0,\frac{1}{2})$ + reality condition	4	2	spin $\frac{1}{2}$, its own antiparticle
Dirac spinor	$(\frac{1}{2},0)\oplus(0,\frac{1}{2})$	8	4	spin-$\frac{1}{2}$ particle and antiparticle
Massless vector A_μ	$(\frac{1}{2},\frac{1}{2})$	$4\to 3^*$	2	$h=\pm 1$
Massive vector	$(\frac{1}{2},\frac{1}{2})$	4	3	three states of spin 1

The numbers in the table all count real degrees of freedom. "Off shell" means the number of real components of the field without using the equations of motion; "on shell" means the number of physical states left after using the equations of motion and removing gauge redundancy. * A massless vector field has 4 components off shell, and 3 after removing the gauge redundancy. Notice that first-order equations (Dirac, Weyl) halve the number of components on shell, because they tie half of the components to the other half.

Why keep the books so carefully? Because supersymmetry turns bosons into fermions and fermions into bosons, so within a single supersymmetry multiplet the bosonic and fermionic degrees of freedom must be equal in number. Look at the simplest combination in the table: a complex scalar (2 on shell) paired with a Weyl spinor (2 on shell) balances exactly on shell; this is the simplest *chiral multiplet*. But what about off shell? The complex scalar has 2, while the Weyl spinor has 4, and the books don't balance. To balance them we must bring in one more non-propagating "auxiliary field". That is a story for the next part.

Problem 7.4 Pair a massless vector field with a Majorana spinor. Are the bosonic and fermionic degrees of freedom equal on shell? By how much do they differ off shell?

Appendix A

Notation and Conventions

This book uses one set of conventions, and only one, from beginning to end. When comparing with other books, check this table first.

Object	Convention used in this book
Units	$\hbar = c = 1$
Metric	$\eta_{\mu\nu} = \mathrm{diag}(1,-1,-1,-1)$; $x_\mu = (t, -\boldsymbol{x})$; timelike vectors have $x^2 > 0$
Indices	$\mu, \nu, \ldots = 0, 1, 2, 3$; $i, j, k = 1, 2, 3$; $\alpha, \beta = 1, 2$ (undotted); $\dot\alpha, \dot\beta = 1, 2$ (dotted)
Three-dimensional ε	$\varepsilon_{123} = 1$
Four-dimensional ε	$\varepsilon^{0123} = -1$, $\varepsilon_{0123} = +1$
Momentum and plane waves	$P_\mu = \mathrm{i}\partial_\mu$; positive-frequency plane wave $e^{-\mathrm{i}p\cdot x}$
Lorentz generators	$L_{\mu\nu} = \mathrm{i}(x_\mu\partial_\nu - x_\nu\partial_\mu)$ (orbital part); $D(\omega) = \exp(-\frac{\mathrm{i}}{2}\omega^{\mu\nu}L_{\mu\nu})$
Rotations and boosts	$R_i = \frac{1}{2}\varepsilon_{ijk}L_{jk}$, $K_i = L^{0i}$; the boost along $+z$ is $e^{-\mathrm{i}\chi K_3}$
Left and right combinations	$J^L = \frac{1}{2}(\boldsymbol{R} + \mathrm{i}\boldsymbol{K})$, $J^R = \frac{1}{2}(\boldsymbol{R} - \mathrm{i}\boldsymbol{K})$
Pauli–Lubanski	$W^\mu = \frac{1}{2}\varepsilon^{\mu\nu\rho\sigma}P_\nu L_{\rho\sigma}$; in the rest frame $\boldsymbol{W} = m\boldsymbol{R}$
σ matrices	$\sigma^\mu = (\mathbf{1}, \boldsymbol{\sigma})$, $\bar\sigma^\mu = (\mathbf{1}, -\boldsymbol{\sigma})$; $\bar\sigma^\mu = \mathrm{adj}\,\sigma^\mu$
Spinor ε	$\varepsilon^{12} = -\varepsilon^{21} = 1$, $\varepsilon_{12} = -\varepsilon_{21} = -1$, same for dotted indices; $\varepsilon^{\alpha\beta}\varepsilon_{\beta\gamma} = \delta^\alpha{}_\gamma$
Raising and lowering spinor indices	$\psi^\alpha = \varepsilon^{\alpha\beta}\psi_\beta$, $\psi_\alpha = \varepsilon_{\alpha\beta}\psi^\beta$ (ε on the left, contracted on its second index)
Spinor contractions	$\psi\chi = \psi^\alpha\chi_\alpha$, $\bar\psi\bar\chi = \bar\psi_{\dot\alpha}\bar\chi^{\dot\alpha}$
Spinor transformations	$\psi_\alpha \to M_\alpha{}^\beta\psi_\beta$, $\bar\psi_{\dot\alpha} \to (M^*)_{\dot\alpha}{}^{\dot\beta}\bar\psi_{\dot\beta}$; $\psi_\alpha \in (\frac{1}{2}, 0)$
$\sigma^{\mu\nu}$	$\sigma^{\mu\nu} = \frac{\mathrm{i}}{4}(\sigma^\mu\bar\sigma^\nu - \sigma^\nu\bar\sigma^\mu)$, $\bar\sigma^{\mu\nu} = \frac{\mathrm{i}}{4}(\bar\sigma^\mu\sigma^\nu - \bar\sigma^\nu\sigma^\mu)$
Self-duality	$\frac{1}{2}\varepsilon^{\mu\nu\rho\sigma}\sigma_{\rho\sigma} = -\mathrm{i}\sigma^{\mu\nu}$, $\frac{1}{2}\varepsilon^{\mu\nu\rho\sigma}\bar\sigma_{\rho\sigma} = +\mathrm{i}\bar\sigma^{\mu\nu}$
γ matrices	$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$; Weyl representation $\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu \\ \bar\sigma^\mu & 0 \end{pmatrix}$
γ^5	$\mathrm{i}\gamma^0\gamma^1\gamma^2\gamma^3 = \mathrm{diag}(-\mathbf{1}, \mathbf{1})$
Dirac spinor	$\Psi = (\psi_\alpha, \bar\chi^{\dot\alpha})^{\mathsf{T}}$, $\bar\Psi = \Psi^\dagger\gamma^0 = (\chi^\alpha, \bar\psi_{\dot\alpha})$; $\Sigma^{\mu\nu} = \frac{\mathrm{i}}{4}[\gamma^\mu, \gamma^\nu]$
Lagrangians	$\mathcal{L}_{\text{Dirac}} = \bar\Psi(\mathrm{i}\gamma^\mu\partial_\mu - M)\Psi$; $\mathcal{L}_{\text{Weyl}} = \mathrm{i}\bar\psi\bar\sigma^\mu\partial_\mu\psi$
Charge conjugation	$\Psi^c = C\bar\Psi^{\mathsf{T}}$, $C = \mathrm{i}\gamma^0\gamma^2 = \mathrm{diag}(\varepsilon_{\alpha\beta}, \varepsilon^{\dot\alpha\dot\beta})$; $\Psi^c = (\chi_\alpha, \bar\psi^{\dot\alpha})^{\mathsf{T}}$
Electromagnetic field	$A^\mu = (\phi, \boldsymbol{A})$; $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$; $F_{0i} = E^i$, $F_{ij} = -\varepsilon_{ijk}B^k$

Appendix B

Solutions to the Problems

Chapter 1

Problem 1.1 $x_\mu = (2,-1,0,0)$ and $x^2 = 4-1 = 3$. Since $x_1 = -x^1$ while $\partial_1 = \partial/\partial x^1$, we have $\partial_1 x_1 = \partial(-x^1)/\partial x^1 = -1 = \eta_{11}$. "Same index" does not mean the result is $+1$; it depends on whether the coordinate being differentiated has an upper or a lower index.

Problem 1.2 $v = 3/5$ gives $\gamma = 5/4$, so $\cosh\chi = 5/4$ and $\sinh\chi = \gamma v = 3/4$. Then

$$t' = \tfrac{5}{4}\cdot 2 + \tfrac{3}{4}\cdot 1 = \tfrac{13}{4}, \qquad z' = \tfrac{3}{4}\cdot 2 + \tfrac{5}{4}\cdot 1 = \tfrac{11}{4}, \qquad t'^2 - z'^2 = \tfrac{169-121}{16} = 3 = t^2 - z^2.$$

Problem 1.3 We need only look at the 2×2 block for t, z:

$$\begin{pmatrix} c_1 & s_1 \\ s_1 & c_1 \end{pmatrix}\begin{pmatrix} c_2 & s_2 \\ s_2 & c_2 \end{pmatrix} = \begin{pmatrix} c_1c_2 + s_1s_2 & c_1s_2 + s_1c_2 \\ s_1c_2 + c_1s_2 & s_1s_2 + c_1c_2 \end{pmatrix} = \begin{pmatrix} \cosh(\chi_1+\chi_2) & \sinh(\chi_1+\chi_2) \\ \sinh(\chi_1+\chi_2) & \cosh(\chi_1+\chi_2) \end{pmatrix},$$

where $c_i = \cosh\chi_i$ and $s_i = \sinh\chi_i$, and the last step uses the addition formulas for hyperbolic functions. So $v = \tanh(\chi_1+\chi_2) = (\tanh\chi_1 + \tanh\chi_2)/(1+\tanh\chi_1\tanh\chi_2) = (v_1+v_2)/(1+v_1v_2)$.

Problem 1.4 Call the inverse (Λ', a'). From $(\Lambda, a)(\Lambda', a') = (\Lambda\Lambda', a + \Lambda a') = (\mathbf{1}, 0)$ we get $\Lambda' = \Lambda^{-1}$ and $a' = -\Lambda^{-1}a$. Multiplying in the other order works too: $(\Lambda^{-1}, -\Lambda^{-1}a)(\Lambda, a) = (\mathbf{1}, -\Lambda^{-1}a + \Lambda^{-1}a) = (\mathbf{1}, 0)$.

Problem 1.5 Write $f_x = \partial_x f$, and so on. Compute in both orders:

$$L_{12}P^1 f = -(xf_{xy} - yf_{xx}), \qquad P^1 L_{12} f = -\partial_x(xf_y - yf_x) = -(f_y + xf_{xy} - yf_{xx}).$$

Subtracting, all the second derivatives cancel, and $[L_{12}, P^1]f = f_y$. Since $P^2 = -P_2 = -\mathrm{i}\partial_y$, we have $f_y = \mathrm{i}P^2 f$, i.e., $[L_{12}, P^1] = \mathrm{i}P^2$. From Eq. (1.29), $[L_{12}, P_1] = \mathrm{i}(\eta_{21}P_1 - \eta_{11}P_2) = \mathrm{i}P_2$, which is the same result once the index is raised.

Chapter 2

Problem 2.1 $[J_1^L, J_2^R] = \frac{1}{4}\big([R_1,R_2] - \mathrm{i}[R_1,K_2] + \mathrm{i}[K_1,R_2] + [K_1,K_2]\big) = \frac{1}{4}\big(\mathrm{i}R_3 + K_3 - K_3 - \mathrm{i}R_3\big) = 0$. The two R_3's cancel, and the two K_3's cancel; the cancellation of the first pair relies on the minus sign in $[K_i, K_j]$.

Problem 2.2 Take $\mu\nu\rho\sigma = 1223$ in Eq. (1.33): $[L_{12}, L_{23}] = \mathrm{i}(\eta_{22}L_{13} - \eta_{12}L_{23} - \eta_{23}L_{12} + \eta_{13}L_{22}) = -\mathrm{i}L_{13} = \mathrm{i}L_{31}$. With $R_3 = L_{12}$, $R_1 = L_{23}$, $R_2 = L_{31}$, this is $[R_3, R_1] = \mathrm{i}R_2$.

Problem 2.3 The dimension is $(2\cdot 1 + 1)(2\cdot\frac{1}{2} + 1) = 6$. Under rotations it decomposes as $1 \otimes \frac{1}{2} = \frac{1}{2} \oplus \frac{3}{2}$, with dimensions $2 + 4 = 6$, which checks.

Problem 2.4 $x_\mu = (3,-1,-2,0)$, so $X = \begin{pmatrix} 3 & -1+2\mathrm{i} \\ -1-2\mathrm{i} & 3 \end{pmatrix}$ and $\det X = 9-5 = 4 = 3^2-1^2-2^2$. Since $\bar\sigma^1 = -\sigma_1$, $\frac12\operatorname{tr}(X\bar\sigma^1) = -\frac12\big[(-1+2\mathrm{i})+(-1-2\mathrm{i})\big] = 1 = x^1$. If you forget the minus sign in $\bar\sigma^1$, what you read off will be x_1.

Problem 2.5 M_R is diagonal with diagonal elements of modulus 1, so the diagonal elements of X are unchanged; the upper right element gets multiplied by $e^{-\mathrm{i}\theta/2}\cdot e^{-\mathrm{i}\theta/2} = e^{-\mathrm{i}\theta}$:

$$-x' + \mathrm{i}y' = e^{-\mathrm{i}\theta}(-x+\mathrm{i}y) = (-x\cos\theta + y\sin\theta) + \mathrm{i}(x\sin\theta + y\cos\theta),$$

i.e., $x' = x\cos\theta - y\sin\theta$ and $y' = x\sin\theta + y\cos\theta$, exactly a rotation about the z axis. When $\theta = 2\pi$, $M_R = -\mathbf{1}$, but $MXM^\dagger = X$: the vector is back where it started, while the spinor changes sign.

Problem 2.6 Apply M_2 first and then M_1: $X \mapsto M_2XM_2^\dagger \mapsto M_1(M_2XM_2^\dagger)M_1^\dagger = (M_1M_2)X(M_1M_2)^\dagger$. The vector transformation corresponding to the left side is $\Lambda(M_2)$ followed by $\Lambda(M_1)$; the right side is $\Lambda(M_1M_2)$. They agree for every x, so $\Lambda(M_1M_2) = \Lambda(M_1)\Lambda(M_2)$.

Chapter 3

Problem 3.1 $C_1 = m^2$ and $C_2 = -m^2\cdot 1\cdot 2 = -2m^2$; at fixed momentum there are three states, $r = -1, 0, 1$. But the momentum $\boldsymbol{p}$ can vary continuously over the whole mass shell, and a general wave packet is described by three momentum-space wave functions $f_{-1}(\boldsymbol{p}), f_0(\boldsymbol{p}), f_1(\boldsymbol{p})$, so the full state space is infinite-dimensional.

Problem 3.2 Using Eq. (3.18): $[N_2, P^0] = -[K_1, P^0] = \mathrm{i}P^1$; $[N_2, P^1] = [R_2, P^1] - [K_1, P^1] = -\mathrm{i}P^3 + \mathrm{i}P^0$; $[N_2, P^2] = 0$; $[N_2, P^3] = [R_2, P^3] = \mathrm{i}P^1$. At $k = (\kappa, 0, 0, \kappa)$ we have $P^1 = 0$ and $P^0 - P^3 = 0$, so all of them vanish.

Problem 3.3 $W^2 = h^2P^2 = 0$. But you cannot conclude from this that $h = 0$: take $h = 1$ and $p^\mu = (\kappa, 0, 0, \kappa)$; then $W^\mu = (\kappa, 0, 0, \kappa)$ is nonzero, yet its length squared is $\kappa^2 - \kappa^2 = 0$. Nor can the photon's two helicities be told apart by P^2 and W^2.

Problem 3.4 From Eq. (3.26), $W^0 = \boldsymbol{p}\cdot\boldsymbol{R} = pR_3$. The momentum is along z, so the helicity is $\hat{\boldsymbol{p}}\cdot\boldsymbol{R} = R_3$, and therefore $W^0/|\boldsymbol{p}| = R_3 = h$.

Chapter 4

Problem 4.1 The variation of the kinetic term is unchanged, and $\partial\mathcal{L}/\partial\phi = -m^2\phi - \lambda\phi^3/3!$, so $(\Box + m^2)\phi + \frac{\lambda}{3!}\phi^3 = 0$. The equation is no longer linear, and the sum of two solutions is in general no longer a solution; in Fourier space ϕ^3 becomes a convolution of three momentum components, so modes of different momenta are no longer independent of one another.

Problem 4.2 $[g_n] + n[\phi] = D$ and $[\phi] = (D-2)/2$, so $[g_n] = D - n(D-2)/2$. For $D = 4$, $[g_n] = 4 - n$: the coupling constant of ϕ^3 has dimension 1 (a mass), and the coupling constant of ϕ^4 is dimensionless.

Problem 4.3

$$\partial_\mu\varphi^*\partial^\mu\varphi = \tfrac12(\partial_\mu\phi_1 - \mathrm{i}\partial_\mu\phi_2)(\partial^\mu\phi_1 + \mathrm{i}\partial^\mu\phi_2) = \tfrac12(\partial\phi_1)^2 + \tfrac12(\partial\phi_2)^2 + \tfrac{\mathrm{i}}{2}\big(\partial_\mu\phi_1\partial^\mu\phi_2 - \partial_\mu\phi_2\partial^\mu\phi_1\big).$$

The last term vanishes because the metric is symmetric. Adding $\varphi^*\varphi = \frac{1}{2}(\phi_1^2 + \phi_2^2)$, we get the sum of the two real-field Lagrangians. Without the $1/\sqrt{2}$, the right side would be $(\partial\phi_1)^2 + \cdots$, missing the standard $\frac{1}{2}$, and the real fields would have to be redefined.

Chapter 5

Problem 5.1 $\psi^\alpha = (b, -a)$ and $\chi^\alpha = (d, -c)$. $\psi^\alpha\chi_\alpha = bc - ad$ and $\psi_\alpha\chi^\alpha = ad - bc$. They differ by a minus sign, and here we haven't swapped the order of any two spinors; the minus sign comes entirely from the antisymmetry of ε.

Problem 5.2 (a) Multiply step by step:

$$\varepsilon A^{\mathsf{T}} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} a & c \\ b & d \end{pmatrix} = \begin{pmatrix} b & d \\ -a & -c \end{pmatrix}, \qquad \begin{pmatrix} b & d \\ -a & -c \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Comparing with Eq. (5.10), $\bar\sigma^\mu = \operatorname{adj}\sigma^\mu$. For example, $\operatorname{adj}\sigma_3 = \operatorname{diag}(-1, 1) = -\sigma_3$.

(b) The adjugate of a 2×2 matrix is a linear operation, so $\operatorname{adj} X = x_\mu\bar\sigma^\mu$. Take the adjugate of both sides of $X' = MXM^\dagger$:

$$x'_\mu\bar\sigma^\mu = \operatorname{adj}(M^\dagger)\,(x_\nu\bar\sigma^\nu)\,\operatorname{adj} M = (M^\dagger)^{-1}(x_\nu\bar\sigma^\nu)M^{-1}.$$

On the left, $x'_\mu = \Lambda_\mu{}^\nu x_\nu$. Comparing the coefficients of x_ν, $(M^\dagger)^{-1}\bar\sigma^\nu M^{-1} = \Lambda_\mu{}^\nu\bar\sigma^\mu$, i.e., $M^\dagger\bar\sigma^\mu M\,\Lambda_\mu{}^\nu = \bar\sigma^\nu$. Since $\Lambda_\mu{}^\nu = (\Lambda^{-1})^\nu{}_\mu$, multiplying both sides by $\Lambda^\rho{}_\nu$ gives $M^\dagger\bar\sigma^\rho M = \Lambda^\rho{}_\nu\bar\sigma^\nu$.

Problem 5.3 $\gamma^0 = \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix}$ and $\gamma^1 = \begin{pmatrix} 0 & \sigma_1 \\ -\sigma_1 & 0 \end{pmatrix}$. So $(\gamma^1)^2 = \operatorname{diag}(-\sigma_1^2, -\sigma_1^2) = -\mathbf{1}$; $\gamma^0\gamma^1 = \operatorname{diag}(-\sigma_1, \sigma_1)$ and $\gamma^1\gamma^0 = \operatorname{diag}(\sigma_1, -\sigma_1)$, which add to zero. This agrees with $2\eta^{11} = -2$ and $2\eta^{01} = 0$.

Problem 5.4 $\gamma^5 = \operatorname{diag}(-\mathbf{1}, \mathbf{1})$, whose square is obviously $\mathbf{1}$. Every γ^μ has the off-diagonal block form $\begin{pmatrix} 0 & A \\ B & 0 \end{pmatrix}$, so $\gamma^5\gamma^\mu = \begin{pmatrix} 0 & -A \\ B & 0 \end{pmatrix}$ and $\gamma^\mu\gamma^5 = \begin{pmatrix} 0 & A \\ -B & 0 \end{pmatrix}$, which add to zero.

Problem 5.5 $\Psi = v\,e^{+\mathrm{i}Mt}$ gives $(-M\gamma^0 - M)v = 0$, i.e., $\gamma^0 v = -v$. γ^0 swaps the upper and lower blocks, so the lower block equals minus the upper block: $v = (\eta, -\eta)^{\mathsf{T}}$, with η an arbitrary two-component column vector. There are two independent solutions.

Problem 5.6 $\boldsymbol{p}\cdot\boldsymbol{\sigma} = E\sigma_1$, and the equation is $E(\mathbf{1} + \sigma_1)u = 0$, i.e., $\sigma_1 u = -u$, with solution $u \propto (1, -1)^{\mathsf{T}}/\sqrt{2}$. The helicity operator is $\hat{\boldsymbol{p}}\cdot\boldsymbol{\sigma}/2 = \sigma_1/2$, which takes the value $-\frac{1}{2}$ on this solution. The conclusion doesn't depend on the direction of motion, only on the chirality of the spinor.

Chapter 6

Problem 6.1 Take $\psi = \chi = \theta$ in the Fierz identity. The symmetric part is proportional to $\theta_\alpha\theta_\beta + \theta_\beta\theta_\alpha$, which vanishes by Grassmann anticommutation, so $\theta_\alpha\theta_\beta = \frac{1}{2}\varepsilon_{\alpha\beta}(\theta\theta)$. Check $\alpha = 1, \beta = 2$: $\theta\theta = 2\theta_2\theta_1$, and the right side is $\frac{1}{2}(-1)(2\theta_2\theta_1) = -\theta_2\theta_1 = \theta_1\theta_2$, the same as the left side.

Problem 6.2 $\frac{1}{2}\varepsilon^{12\rho\sigma}\sigma_{\rho\sigma} = \varepsilon^{1203}\sigma_{03}$ (the terms with $\rho\sigma = 03$ and 30 are equal). 1203 is an even permutation of 0123, so $\varepsilon^{1203} = \varepsilon^{0123} = -1$; and $\sigma_{03} = -\sigma^{03} = \frac{\mathrm{i}}{2}\sigma_3$. The left side is $-\frac{\mathrm{i}}{2}\sigma_3$. The right side is $-\mathrm{i}\sigma^{12} = -\mathrm{i}\cdot\frac{1}{2}\sigma_3 = -\frac{\mathrm{i}}{2}\sigma_3$. They are equal.

Problem 6.3 Charge conjugation swaps (ψ, χ) into (χ, ψ), and doing it again swaps them back, so $(\Psi^c)^c = \Psi$. In the equation satisfied by Ψ^c the charge is $-e$. If we demand $\Psi^c = \Psi$, the same field must carry both charge e and charge $-e$, which is possible only if $e = 0$.

Problem 6.4 Set $\chi = \psi$; then $\bar{\Psi}_M = (\psi^\alpha, \bar{\psi}_{\dot\alpha})$, and Eq. (5.32) gives

$$\bar{\Psi}_M\,\mathrm{i}\gamma^\mu\partial_\mu\Psi_M = \mathrm{i}\bar\psi\bar\sigma^\mu\partial_\mu\psi + \mathrm{i}\psi\sigma^\mu\partial_\mu\bar\psi, \qquad \bar{\Psi}_M\Psi_M = \psi\psi + \bar\psi\bar\psi.$$

By Eq. (6.6), $\mathrm{i}\psi\sigma^\mu\partial_\mu\bar\psi = \mathrm{i}\bar\psi\bar\sigma^\mu\partial_\mu\psi$ plus a total derivative. Multiplying by $\frac{1}{2}$, the kinetic terms combine into a single $\mathrm{i}\bar\psi\bar\sigma^\mu\partial_\mu\psi$, and the mass term is $-\frac{M}{2}(\psi\psi + \bar\psi\bar\psi)$.

Chapter 7

Problem 7.1 $k\cdot\epsilon_\pm = \kappa\cdot 0 - \kappa\cdot 0 = 0$, because the inner product involves only the 0 and 3 components. If $\epsilon^\mu = c\,k^\mu$, then $A_\mu = c\,k_\mu e^{-\mathrm{i}k\cdot x} = \partial_\mu\big(\mathrm{i}c\,e^{-\mathrm{i}k\cdot x}\big)$ is pure gauge, and the field strength $F_{\mu\nu} \propto k_\mu k_\nu - k_\nu k_\mu = 0$: it produces no electric or magnetic field whatsoever.

Problem 7.2 $F_{\mu\nu}F^{\mu\nu} = 2F_{0i}F^{0i} + F_{ij}F^{ij}$. Since $F^{0i} = -F_{0i} = -E^i$ and $F^{ij} = F_{ij}$, the first term is $-2\boldsymbol{E}^2$; the second is $\varepsilon_{ijk}\varepsilon_{ijl}B^kB^l = 2\boldsymbol{B}^2$. So $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} = \frac{1}{2}(\boldsymbol{E}^2 - \boldsymbol{B}^2)$.

Problem 7.3 The constraint $\partial_\mu A^\mu = 0$ gives $k\cdot\epsilon = E\epsilon^0 - k\epsilon^3 = 0$, leaving three independent polarizations: the two transverse ones $\epsilon_\pm = (0, 1, \pm\mathrm{i}, 0)/\sqrt{2}$, and a longitudinal one $\epsilon_L = (k, 0, 0, E)/m$ (you can check that $k\cdot\epsilon_L = 0$ and $\epsilon_L^2 = -1$). The mass term breaks gauge symmetry, so there is no longer an $\epsilon \to \epsilon + ck$ freedom that could be used to remove one more; besides, $k^2 = m^2 \neq 0$, so $\epsilon \propto k$ doesn't even satisfy the constraint.

Problem 7.4 On shell: the vector field has 2 and the Majorana spinor has 2; they are equal. Off shell: the vector field has 3 after removing the gauge redundancy, while the Majorana spinor has 4, so the bosons are one short. Adding one real auxiliary scalar field balances the books, and this is precisely the structure of the supersymmetric vector multiplet.

Appendix C

Further Reading

The following materials can be read alongside this book. Each of them may use conventions different from ours, so keep Appendix A at hand as you read.

D. Tong's lecture notes *Supersymmetric Field Theory* (https://www.damtp.cam.ac.uk/user/tong/susy/susy.pdf) are the natural next step; their first few sections cover the spinors and particle representations of this book and then go straight on to the supersymmetry algebra. By the same author, Chapter 1 of *Quantum Field Theory* (https://www.damtp.cam.ac.uk/user/tong/qft/qfthtml/S1.html) covers classical field theory and the variational principle, and Chapter 1 of *The Standard Model* (https://www.damtp.cam.ac.uk/user/tong/sm/standardmodel1.pdf) has a very clear discussion of the representations of the Poincaré group and the massless little group.

If you want to get to the bottom of the classification of particles, Chapter 2 of S. Weinberg's *The Quantum Theory of Fields*, Vol. I, is the classic; every detail of the little group method is there. Chapter 3 of M. Peskin and D. Schroeder's *An Introduction to Quantum Field Theory* treats the Dirac field, and its notational conventions for the Lorentz group are the same as ours.

The rules for computing with two-component spinors are numerous and easy to get wrong. The long article by H. Dreiner, H. Haber and S. Martin, *Two-component spinor techniques and Feynman rules for quantum field theory and supersymmetry* (arXiv:0812.1594), collects almost every identity you will ever need, and gives formulas for both metric conventions. J. Wess and J. Bagger's *Supersymmetry and Supergravity* is the source of the two-component notation for supersymmetry, though it uses the $(-,+,+,+)$ metric.